

Crop and Soil Data Analysis

I. Dataset Overview

This dataset provides valuable insights into key aspects of agriculture, with a focus on soil type characteristics, humidity, temperature, and moisture levels. It also includes essential nutrient metrics such as nitrogen, potassium, phosphorus, and the corresponding fertilizer types. **The objective is to explore and deeply understand the dataset in order to uncover meaningful patterns and relationships that can support data-driven agricultural decisions.**

Key Features:

1. Temperature

Description: Represents the range of temperatures (in °C) suitable for crop growth, as it impacts germination and plant development.

2. Humidity

Description: Represents the percentage of atmospheric moisture, influencing plant water needs and drought resistance.

3. Moisture

Description: Indicates the percentage of water content in the soil, essential for crop hydration and root absorption.

4. Soil Type

Description: Categorizes soil based on texture and composition (e.g., clayey, black, or other), which affects water retention and nutrient availability.

5. Crop Type

Description: Specifies the type of crop being recommended or grown, categorized by the plant's purpose or growth pattern.

6. Nitrogen

Description: Represents the nitrogen content in the soil (in mg/kg or percentage), critical for leafy growth and overall plant health.

7. Potassium

Description: Represents the potassium content in the soil (in mg/kg or percentage), essential for plant immunity and fruit quality

8. Phosphorus

Description: Represents the phosphorus content in the soil (in mg/kg or percentage), necessary for root development and flowering.

9. Fertilizer Name

Description: Specifies the recommended type of fertilizer to optimize crop growth based on soil deficiencies.

Objectif:

The primary objective is to analyze the relationships between these variables, **identify the Best Fertilizer Based on Soil & Crop Conditions.**

II. Data Cleaning & Preparation

Fix incorrect format:

- In fertilizer name column we have a fertilizer name 10-26-26 show as a format date 10/26/2026, we fix it using find and replace, we replace the wrong format with the correct and change format to general.
- Ensure numerical columns (Temp, Humidity, N, P, K) are in **Number format.**
- For duplicate and missing value we don't have (Remove duplicate using Data Tools → Remove Duplicate & for blanks I use go to special → Blanks)
- I verified and standardize soil and crop name.

III. Initial Exploration

1. Descriptive statistics

This table provides a quick overview of the central tendencies and spread of each feature, helping to highlight patterns such as skewed distributions, variability, and potential nutrient imbalances across the dataset.

	<i>Temperature</i>	<i>Humidity</i>	<i>Moisture</i>	<i>Nitrogen</i>	<i>Potassium</i>	<i>Phosphorous</i>
Mean	30.338895	59.21073125	43.5808625	18.429125	3.916375	18.5125
Standard Error	0.050068491	0.091425729	0.140829303	0.132513932	0.061433812	0.148073681
Median	30.24	59.11	42.25	14	1	18
Mode	40	60	70	10	0	0
Standard Deviation	4.478262019	8.177365841	12.59615583	11.85240637	5.494807182	13.24411264
Sample Variance	20.05483071	66.86931209	158.6631417	140.4795367	30.19290597	175.4065196
	-					
Kurtosis	0.589224438	-0.562918586	-0.752446948	-0.707480372	1.300815762	-1.127571829
Skewness	0.101795335	0.056431713	0.256750919	0.749592463	1.516979576	0.161363174

Range	20	39.98	50	46	23	46
Minimum	20	40.02	20	0	0	0
Maximum	40	80	70	46	23	46
Sum	242711.16	473685.85	348646.9	147433	31331	148100
Count	8000	8000	8000	8000	8000	8000

These statistics provide a good initial understanding of the numerical variables in dataset.

- **General**

Count: Confirms that all numerical variables have 8000 entries.

- **Temperature**

Mean (30.33) / Median (30.24):

Suggests the dataset represents fairly symmetrical distribution with minimal skewness, indicating that the temperature values are evenly distributed without significant outliers or extreme variations.

Range (20 - 40):

Indicates a moderate spread in temperature values, suggesting variability in the dataset but within a controlled range. This range may reflect seasonal or environmental influences on temperature fluctuations.

- **Humidity**

Mean (59.21) / Median (59.11):

The mean and median are very close, indicating a symmetrical distribution with minimal skewness.

Range (40.02 - 80):

A moderate to wide range, suggesting variability in humidity levels, likely due to environmental conditions.

- **Moisture**

Mean (43.5) / Median (42.25):

The mean and median are close, suggesting a fairly normal distribution without strong skewness.

Range (20 - 70):

A broad range, indicating significant variations in soil moisture content, possibly due to differences in irrigation, soil type, or seasonal factors. The variation suggests that the dataset includes both dry and well-irrigated soils.

- **Nitrogen**

Mean (18.42) / Median (14):

The mean is higher than the median, suggesting a right-skewed distribution (some high nitrogen values are pulling the average up).

Range (0 - 46):

A wide range, indicating significant variability in nitrogen levels, likely influenced by soil type, fertilization, and crop usage. While most nitrogen values are lower (as seen in the median), some high nitrogen readings may be affecting the mean, indicating areas with excess fertilization or naturally rich soil.

- **Potassium**

Mean (3.92) / Median (1):

The mean is significantly higher than the median, indicating a strong right-skew (most values are low, but a few very high values pull up the average).

Range (0 - 23):

The dataset includes very low potassium values and a few high concentrations, suggesting localized high-potassium areas.

- **Phosphorous**

Mean (18.51) / Median (18):

The mean and median are almost identical, indicating a symmetrical distribution with no significant skewness.

Range (0 - 46):

A broad range, meaning some soils have very low phosphorus while others have high levels.

2. Feature Categorization

We can categorize the features into four main groups:

- **Soil Characteristics** (Physical & Chemical Properties)
 - Soil Type (Categorical: Clay, Sandy, Loamy, etc.)
 - Moisture (%) (How much water the soil holds)
 - Temperature (°C) (Soil temperature affects crop growth)
- **Environmental Conditions** (Affecting Plant Growth)
 - Humidity → (%) (Air moisture levels)
- **Crop Characteristics (Plant-Specific Factors)**

Crop Type → (Categorical: Wheat, Corn, Rice, etc.)

- **Fertilizer Characteristics (Nutrient Composition)**

Nitrogen (N) → (mg/kg) (Essential for leafy growth)

Phosphorous (P) → (mg/kg) (Helps root development)

Potassium (K) → (mg/kg) (Boosts overall plant health & resistance)

Fertilizer Name → (Categorical: Urea, DAP, NPK, Organic, etc.)

Numerical Columns; Temperature, Humidity, Moisture, Nitrogen, Potassium, Phosphorous

Categorical Columns, Soil Type, Crop Type, Fertilizer Name

3. Unique Value in Soil, Crop and Fertilizer name Columns:

Row Labels	Count of Soil Type	Row Labels	Count of Crop Type	Row Labels	Count of Fertilizer Name
Black	1613	Barley	703	10-26-26	1128
Clayey	1623	Cotton	722	14-35-14	1188
Loamy	1590	Ground Nuts	732	17-17-17	1124
Red	1594	Maize	753	20-20	1103
Sandy	1580	Millet	718	28-28	1120
		Oil seeds	711	DAP	1167
		Paddy	706	Urea	1170
		Pulses	728		
		Sugarcane	763		
		Tobacco	717		
		Wheat	747		

Crop Type: 11 Unique Crop type (Barley, Cotton, Ground Nuts, Maize, Millets, Oil seeds, Paddy, Pulses, Sugarcane, Tobacco and Wheat)

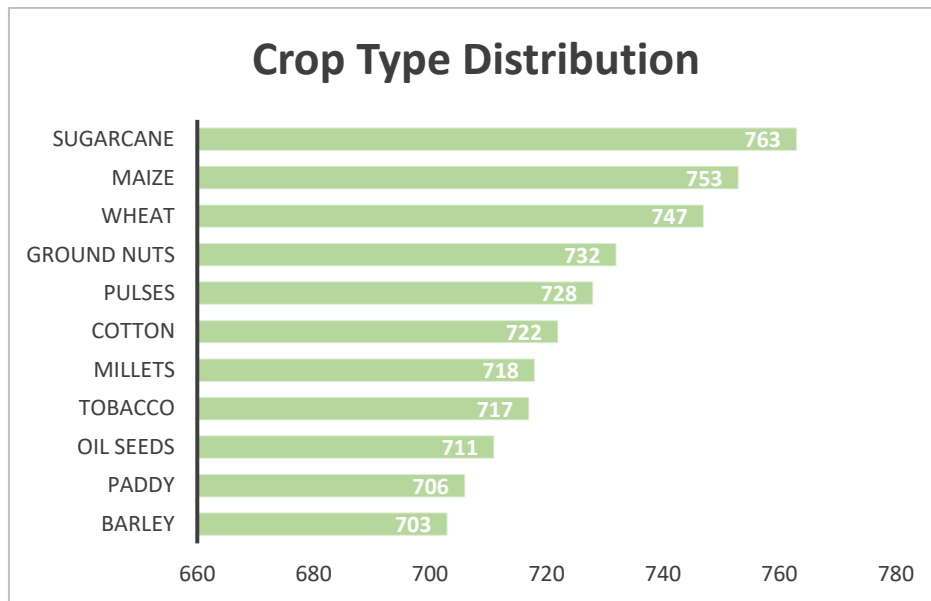
Soil Type: 5 Unique Soil Type (Black, Clayey, Loamy, Red and Sandy)

Fertilizer Names: 7 Unique Fertilizer names (10-26-26, 14-35-14, 17-17-17, 20-20 ,28-28, DAP and Urea)

IV. Exploratory Data Analysis (EDA) in Excel

a. Crop Characteristics

What is the most common crop type reported in the dataset?



Most Common crop type:

- Sugarcane 763: The most frequently reported crop, indicating high demand and widespread cultivation. Sugarcane is often grown in areas with adequate water supply and warm climates, making it a dominant crop in regions with favorable conditions.
- Maize 753: The second most common crop, suggesting its versatility and essential role in food and fodder production. Maize is adaptable to various climates, making it a staple for both human consumption and livestock feed.
- Wheat 747: The third most reported crop, emphasizing its importance as a staple food grain. Wheat cultivation is common in temperate climates and is a major source of carbohydrates for many populations.

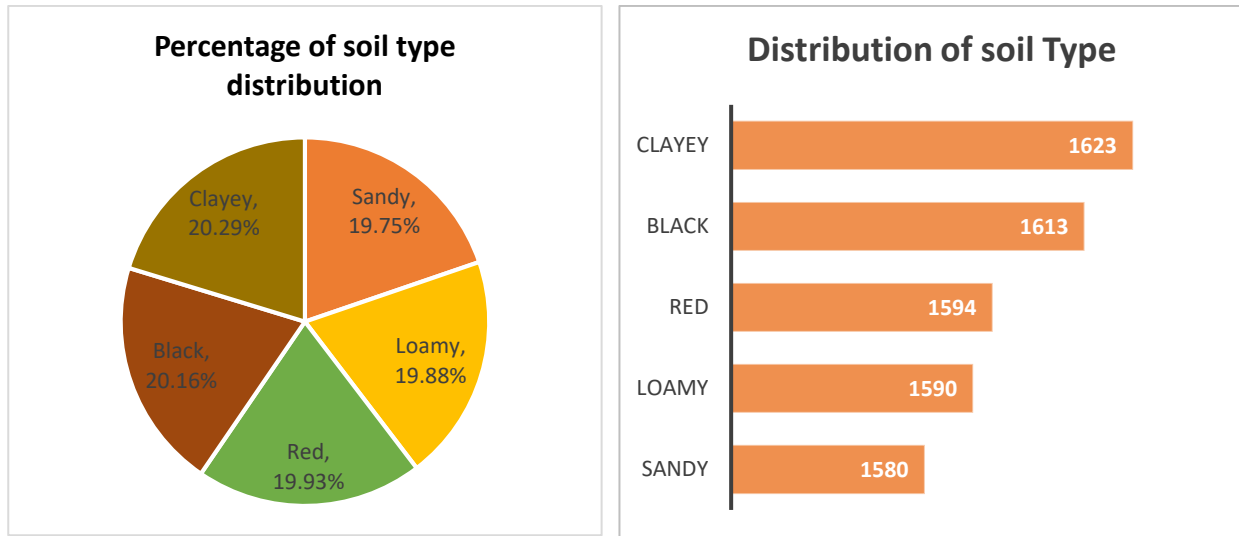
Another Notable crop type:

- Ground Nuts 732
- Pulses 728
- Cotton 722
- Millets 718
- Tobacco 717
- Oil Seeds 711
- Paddy 706
- Barley 703

The analysis reveals that Cereals (Wheat, Maize, Paddy, Barley, Millets) dominate, confirming their staple food role in diets. Cash crops (Cotton, Tobacco, Sugarcane) indicate strong market-driven agriculture, highlighting economic importance. Leguminous crops (Pulses, Ground Nuts) support soil health, making them vital for sustainable farming. Oilseed crops remain crucial for edible oil production, suggesting high domestic and industrial demand.

b. Soil Characteristics

What is the most common soil type reported in the dataset?



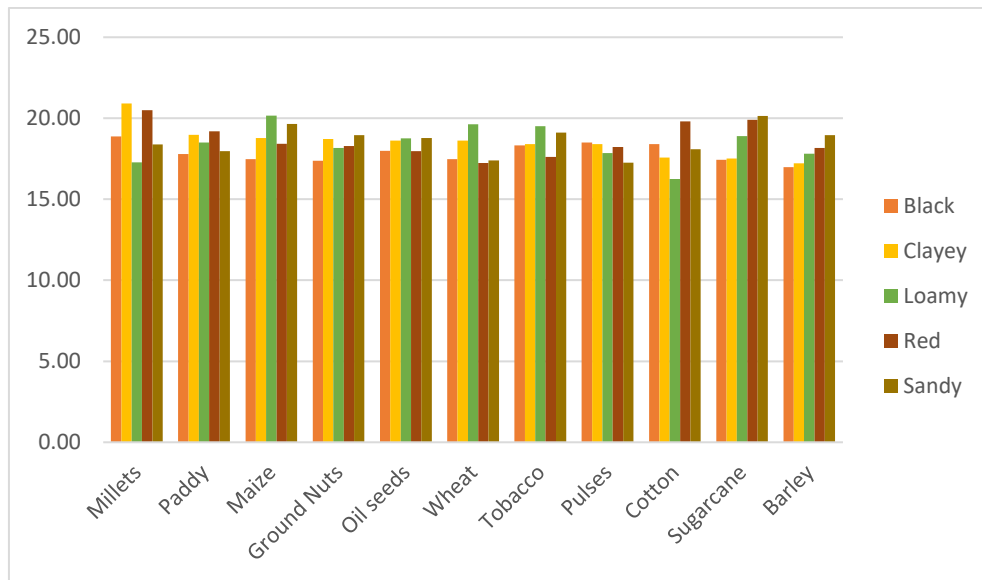
1. Soil Type Distribution:

- Clayey: This is the most prevalent soil type, with 1623 (20.29%) falling into this range.
- Black: The second most common soil type, with 1613 (20.16%).
- Red: is the third soil type with 1594 (19.93%)
- Loamy: 1590 soil type for loamy
- Sandy: the last soil type with 1580

The relatively even distribution of soil types highlights diverse farming conditions, where different soils support different crops. The dominance of clayey and black soils suggests high moisture retention and mineral-rich lands, while the presence of red, loamy, and sandy soils indicates agricultural adaptability and crop diversification strategies.

2. Soil distribution by crop type

How do crop type vary across different soil type?



Millets: Thrives in clayey, red, and black soils.

Paddy: Prefers red, clayey, and loamy soils; least common in black soil.

Maize: Grows best in loamy and sandy soils and less in black soil.

Ground nuts: Mostly in sandy and clayey soils, least in black soil

Oil Seeds: Similar to groundnuts, thriving in sandy and loamy soils.

Wheat: Prefers loamy and clayey soils, grows least in red soil.

Tobacco: Thrives in loamy and sandy soils, least in red soil.

Pulse: Primarily found in black and clayey soils, less in sandy soils.

Cotton: Prefers red and black soils, grows least in loamy soil.

Sugarcane& Barley: Both favor sandy and red soils, grow least in black soil.

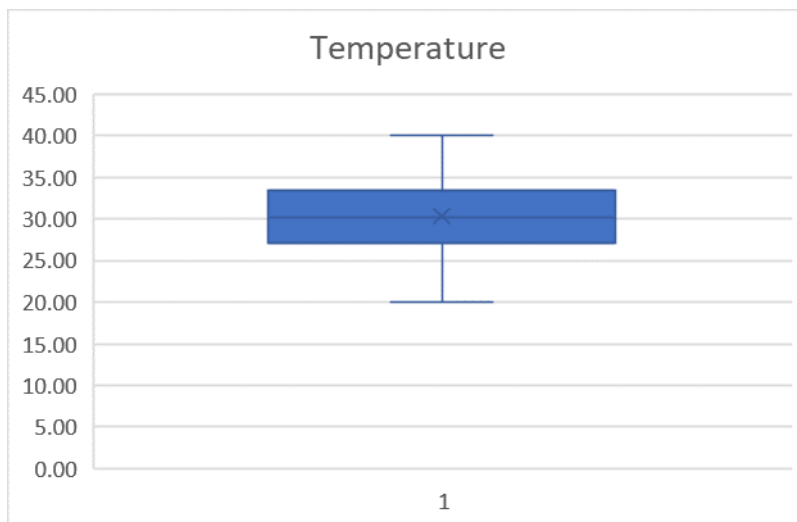
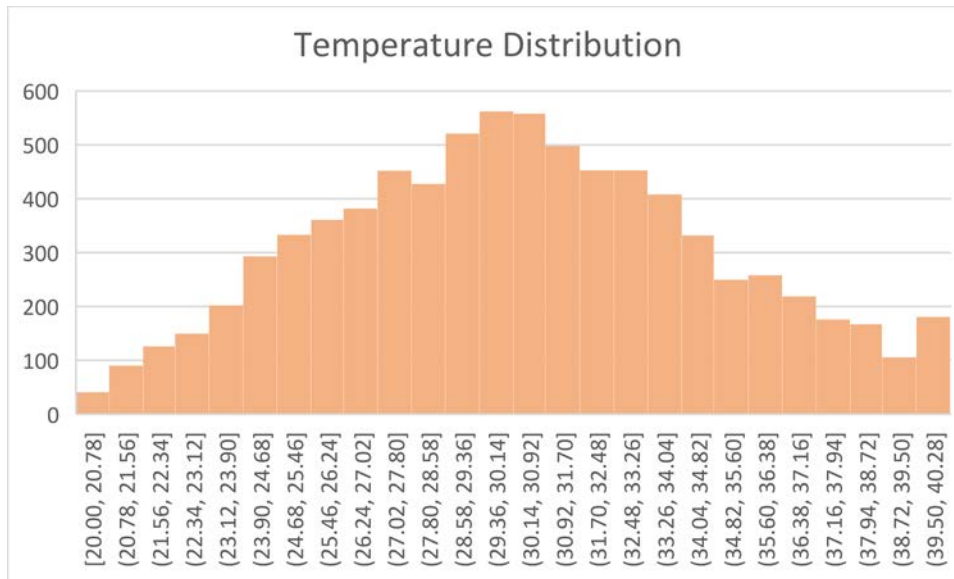
The analysis suggests that:

- **Loamy soil** is the most versatile, supporting a wide range of crops such as maize, wheat, and tobacco. It offers excellent water retention while maintaining good drainage. Known for its richness in nutrients and organic matter, loamy soil is also easy to work with, making it ideal for cultivation.
- **Clayey soil** is commonly associated with millet cultivation. It has excellent water retention but poor drainage. When dry, it becomes very hard and tends to crack.
- **Black soil** plays a vital role in the cultivation of pulses and cotton due to its high nutrient content and excellent moisture retention.
- **Sandy soil** is predominantly used for crops like sugarcane and barley. It offers excellent drainage but poor water retention, is low in nutrients, and although easy to work with, it requires frequent watering and fertilization.

- **Red soil** is well-suited for hardy crops, especially cotton. Although it has lower fertility than black soil, it responds well to fertilizers and has a moderate capacity for water retention.

3. Temperature

What is the average Temperature in the sample?



- **Distribution of temperature:**

The histogram shows the distribution of temperature in the sample. The distribution appears to be normal with a peak around 28-30°C. This suggests that a majority of soil type in the sample in the same zone geographic

- **Box Plot:**

The box plot provides a visual summary of the distribution. The box represents the interquartile range (IQR), which contains 50% of the data.

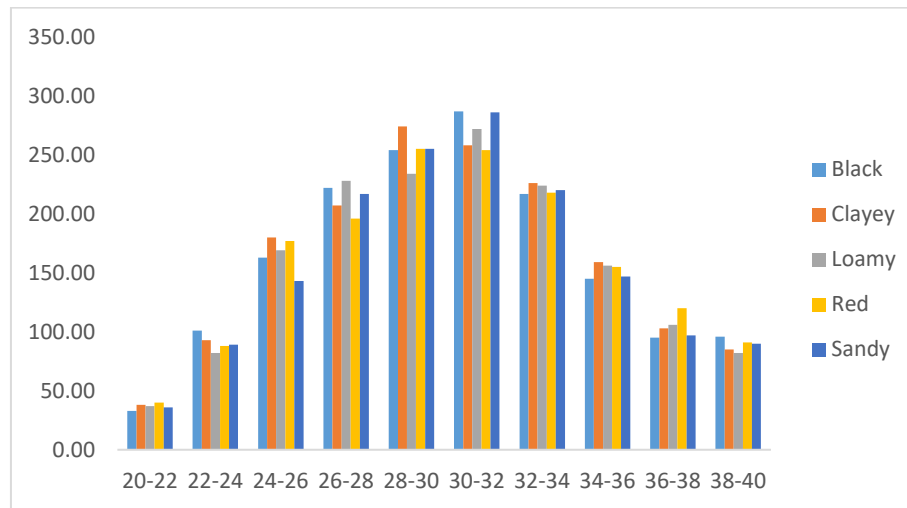
The median temperature appears to be around 30°C, which aligns with the peak of the histogram.

- **Average temperature:**

The average is 30.33°C , suggests that the region under study experiences a **warm climate**, which could be a key factor influencing soil properties and crop suitability.

In summary, based on the histogram, box plot, and the average temperature, the combination of these visualizations confirms that the sampled soil types are from a geographically similar warm region with relatively stable temperature conditions.

Are there significant differences in Temperature distributions between Soil Type?



This histogram represents the **temperature distribution across different soil types** (Black, Clayey, Loamy, Red, and Sandy).

Key Observations:

1. Normal Distribution Shape

- o The data follows a bell-shaped curve, indicating that most temperature values fall within a moderate range (28-32°C), while extreme temperatures are less frequent.

2. Peak Temperature Range (28-32°C)

- o The highest frequency of observations is found in the 28-30°C and 30-32°C bins, meaning that most soil samples experience temperatures within this range.
- o This suggests that the majority of the study area is located in a region with a warm climate.

3. Soil Type Comparison

- o All soil types (Black, Clayey, Loamy, Red, Sandy) follow a similar pattern, indicating that temperature distribution does not vary significantly across soil types.
- o However, Clayey and Black soils tend to have slightly higher counts in the 28-30°C range, suggesting they might be more dominant in regions with these temperatures.

4. Lower Frequency at Extremes (20-22°C & 38-40°C)

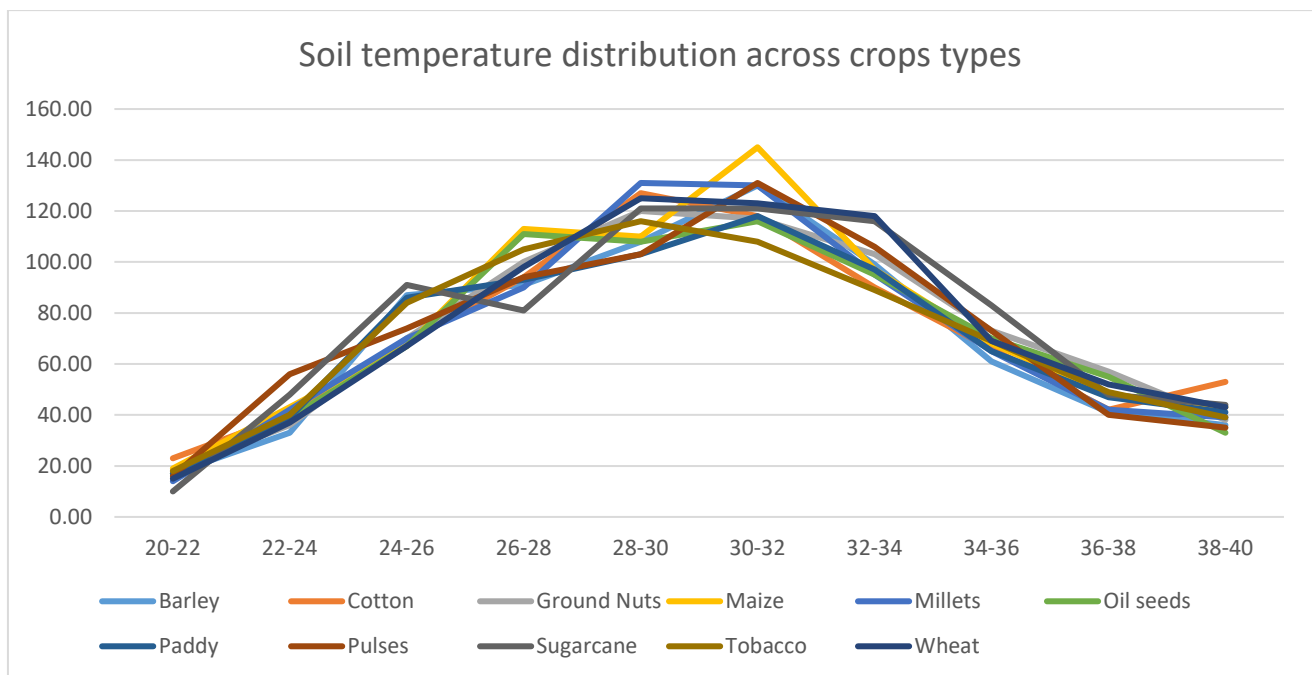
- o Very few soil samples are found in the coldest (20-22°C) and hottest (38-40°C) ranges, confirming that extreme temperatures are less common in the dataset.

The average Temperature between soil types:

Row Labels	Average of Temperature
Red	30.41
Sandy	30.37
Loamy	30.36
Black	30.30
Clayey	30.26

The average temperature across soil types shows minimal variation, consistently hovering around 30°C, with a slight elevation observed in red soil.

Are there significant differences in Temperature distributions between crop Type?



This graph shows the soil temperature distribution across different crop types:

All crops have bell-shaped temperature distribution curves, most crops peak in 28 – 32 °C range showing generally the optimal growing temperature.

Maize appears to have the highest peak at 30-32°C range, and cotton show a unique pattern with relatively higher tolerance to for higher temperature 38 – 40 °C.

All crops show reduced growth potential at the temperature extremes below 22°C or above 38°C.

The average of the soil temperature between crop types:

Row Labels	Average of Temperature
Wheat	30.59
Sugarcane	30.49
Ground Nuts	30.47
Cotton	30.35
Oil seeds	30.33
Paddy	30.33
Maize	30.32
Millets	30.29
Pulses	30.20
Barley	30.20
Tobacco	30.14

The average soil temperature remains relatively consistent across crops, around 30°C, with a slight increase observed in wheat.

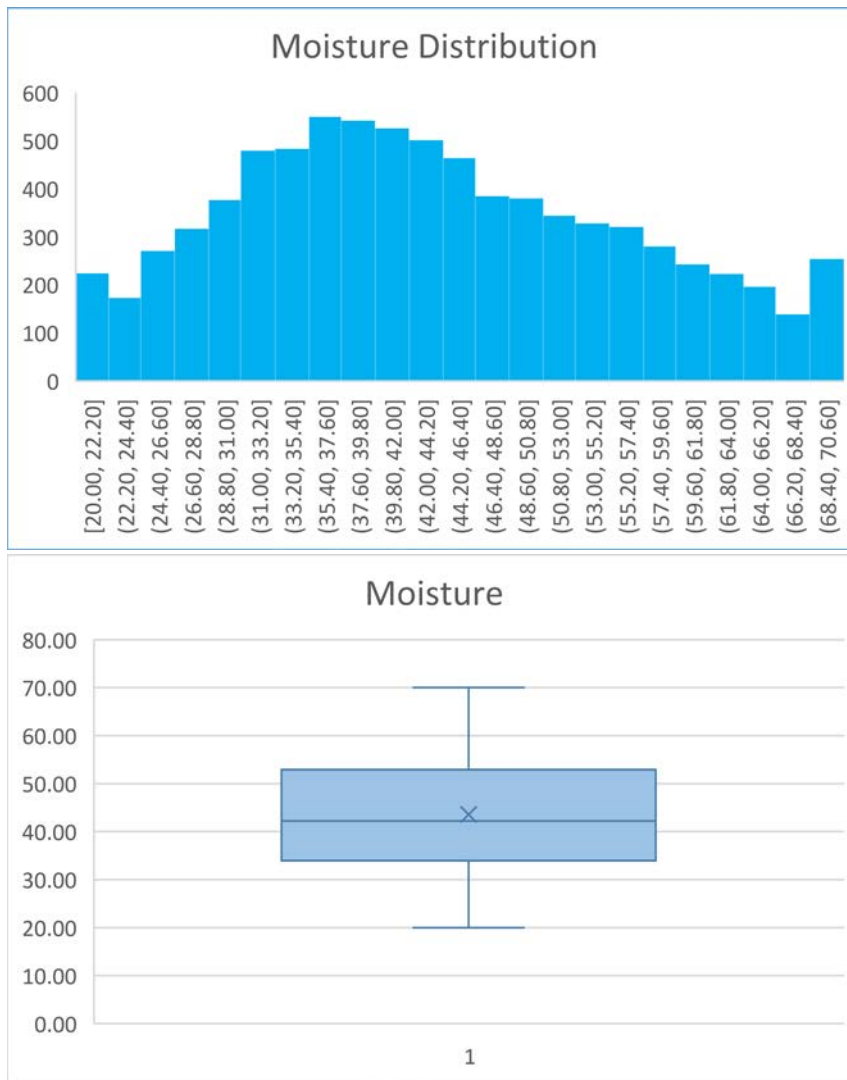
The analysis shows that soil temperature is fairly consistent across all crop types, averaging around 30°C, which falls within the optimal range (28–32°C) for crop development. This consistency suggests that the growing environment across the dataset is relatively uniform and well-suited for a wide variety of crops.

4. Moisture

Soil moisture is the water stored in the soil and is affected by precipitation, temperature, soil characteristics, irrigation and more. These same factors help determine the type of biome present, and the suitability of land for growing crops.

Observing and understanding the consequences of soil moisture fluctuations in crops' root zone will enhance the efforts to maximize yield, improve crop health, and use precious water resources efficiently.

What is the average soil moisture content in the dataset?



- **Distribution of moisture**

This histogram visualizes the distribution of moisture levels.

The distribution is roughly bell-shaped with a slight right skew. It starts low (at 20–24%), rises to a peak between 35% and 42%, then gradually tapers off after 42%.

Most samples have moisture values between 35% and 42%. Additionally, there is a small bump at the end (68–70%), which may indicate a few outliers or a secondary small cluster of samples with high moisture levels.

We can conclude that most soils in the dataset likely belong to the same geographic zone or share a similar type of irrigation.

- **Box Plot:**

The box plot provides a clear visual summary of the moisture distribution.

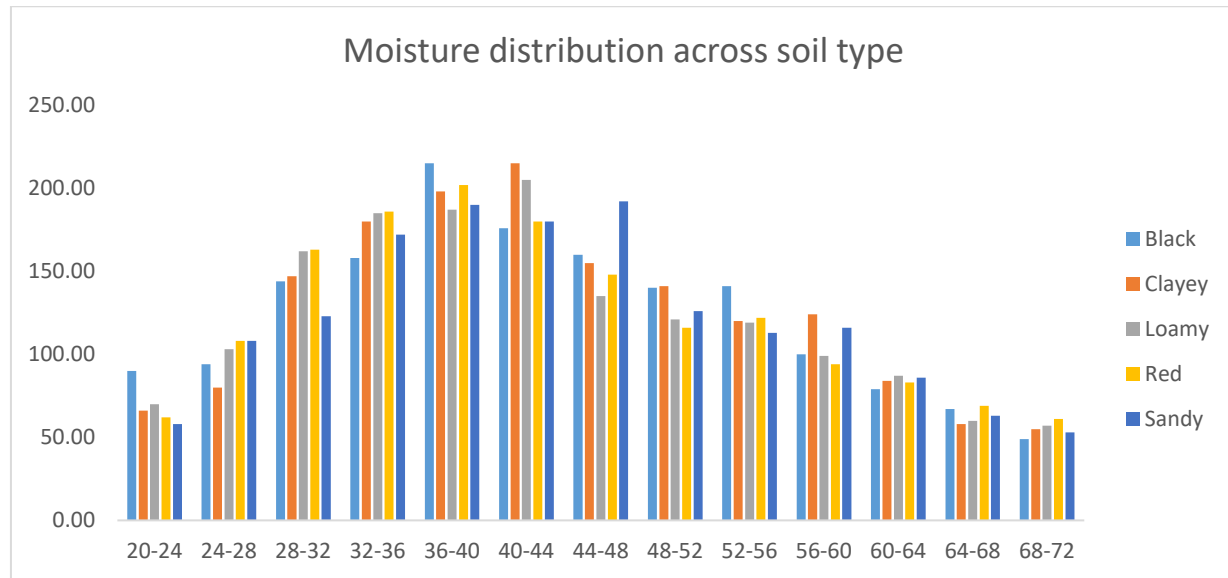
Most moisture values fall between 33% and 55%, indicating a relatively concentrated range. The median is slightly lower than the mean, suggesting a slight right skew, which aligns with the pattern observed in the histogram. The data appears to be evenly distributed, and no extreme outliers are present.

- **Average Moisture:**

The mean of **42.25%** is consistent with the visual distribution seen in both the histogram and the box plot.

In summary, based on the graphs and the average moisture level, we can conclude that the data is concentrated around similar values, suggesting that the soil samples likely come from the same geographic zone or share a uniform irrigation system.

Do moisture levels vary significantly across different soil types?



Row Labels	Average of Moisture
Sandy	43.95
Clayey	43.89
Black	43.42
Red	43.41
Loamy	43.23

- Distribution of moisture across different soil type**

The graph displays the distribution of moisture levels across different soil types. We observe a clear peak between the 36% and 44% moisture range, indicating this is where most samples fall. In contrast, fewer samples are found at the extremes, specifically in the 20–24% and 68–72% ranges.

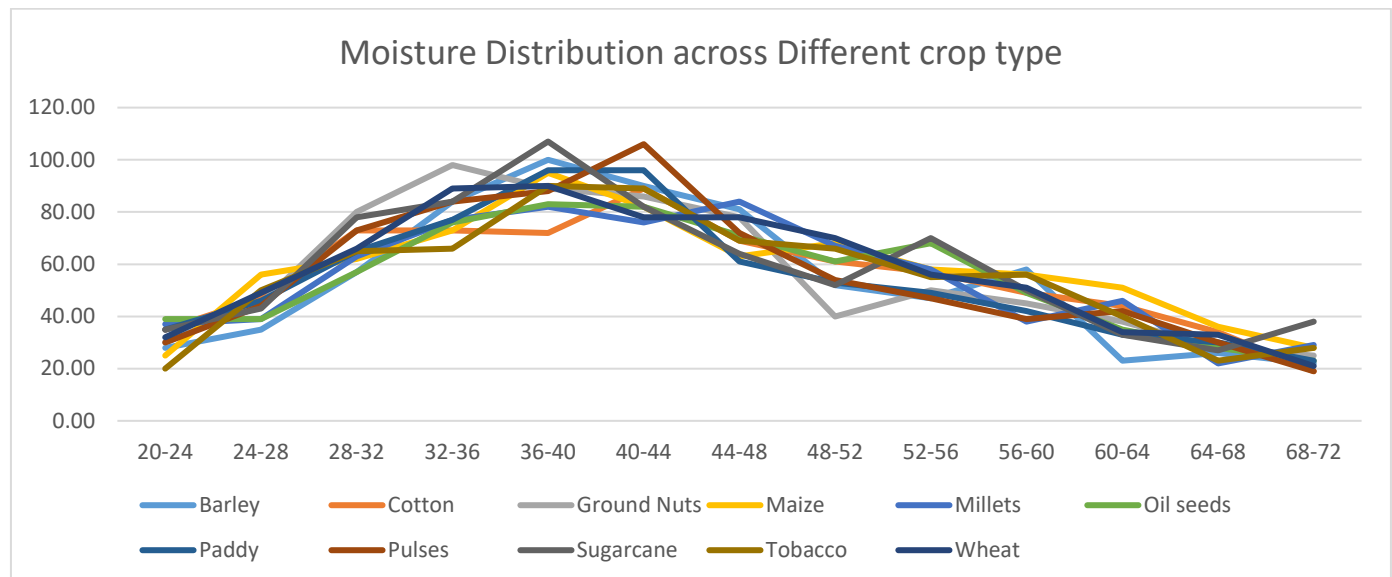
All soil types generally follow a similar distribution shape, suggesting a shared moisture pattern. However, Black and Clayey soils appear more frequently in the 36–44% range, while Sandy soil is slightly more dominant in the 44–48% range. Notably, Black soil also shows higher frequency in the 20–24% range, and Red soil stands out slightly in the 68–72% range.

- **Average moisture for each soil type**

All averages are very close, ranging between 43.23% and 43.95%, this shows little variation across soil types. Slightly, Sandy soil has the highest average moisture.

When we combine the graphs and data, a clear pattern emerges: moisture levels across all soil types are notably consistent, with average values clustering around 43%. While the overall trend is similar among soil types, subtle differences are evident; for instance, sandy soils tend to show slightly higher moisture levels. This may indicate irrigation adjustments made to compensate for sandy soil's naturally low water-holding capacity. These findings suggest that the fields are either located within the same geographic zone or are managed under a uniform irrigation system, resulting in stable moisture levels across the different soil types.

Do moisture levels vary significantly across different crop types?



Row Labels	Average of Moisture
Maize	44.53
Tobacco	44.26
Oil seeds	43.86
Millets	43.79
Cotton	43.65
Sugarcane	43.55
Barley	43.48
Wheat	43.42
Pulses	43.02
Paddy	42.95
Ground Nuts	42.85

- **Distribution of moisture across different crop type**

The graph shows the distribution of moisture levels across different crop types. Most crops follow a similar distribution pattern, with a noticeable peak in the 36–44% range, and fewer occurrences at the extremes (20–24% and 68–72%).

Sugarcane and pulses are more concentrated within the 36–44% range, indicating optimal moisture conditions for their growth. Additionally, sugarcane and oil seeds are more dominant in the 52–56% range, suggesting their tolerance or preference for higher moisture levels. Notably, sugarcane also appears more frequently in the 68–72% range, possibly reflecting its higher water requirements compared to other crops.

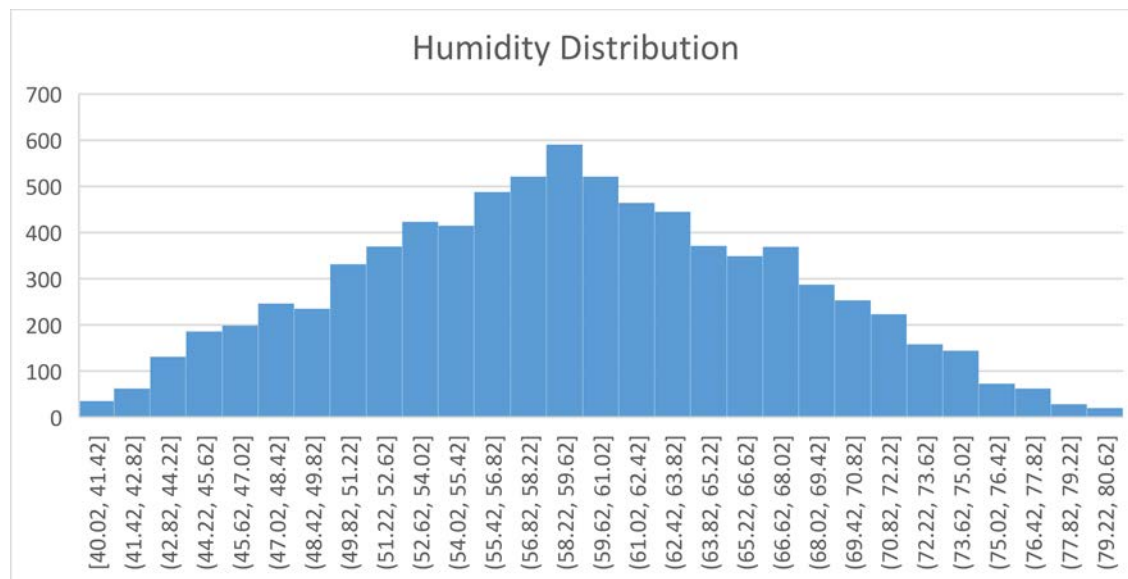
- **Average moisture for each crop type**

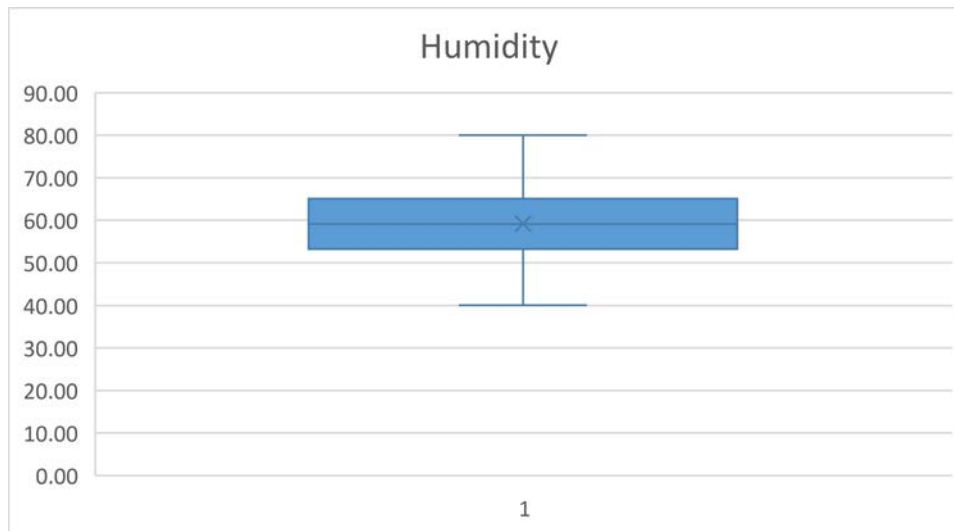
All average moisture values are very close, ranging from 42.85% to 44.53%, indicating minimal variation across different crop types. Among them, maize shows the highest average moisture level, though the difference remains slight.

The narrow range in average moisture values (42.85%–44.53%) across crop types suggests that the environmental or management conditions are highly controlled or consistent.

c. Environmental Conditions

What is the average Humidity in the sample?





- **Humidity Distribution (Histogram):**

The histogram illustrates the distribution of humidity across the dataset. The shape is approximately bell-shaped, beginning with lower values around 40–41%, rising to a peak between 58.22% and 59.62%, and gradually declining after 80%.

The majority of samples fall within the 58% to 59% range, indicating a strong central tendency. This pattern suggests that most soils likely originate from a common geographic region or are subject to similar environmental or management conditions.

- **Box Plot Interpretation:**

The box plot offers a concise visual summary of the humidity data. Most values are concentrated between 53% and 65%, showing a relatively narrow and consistent spread.

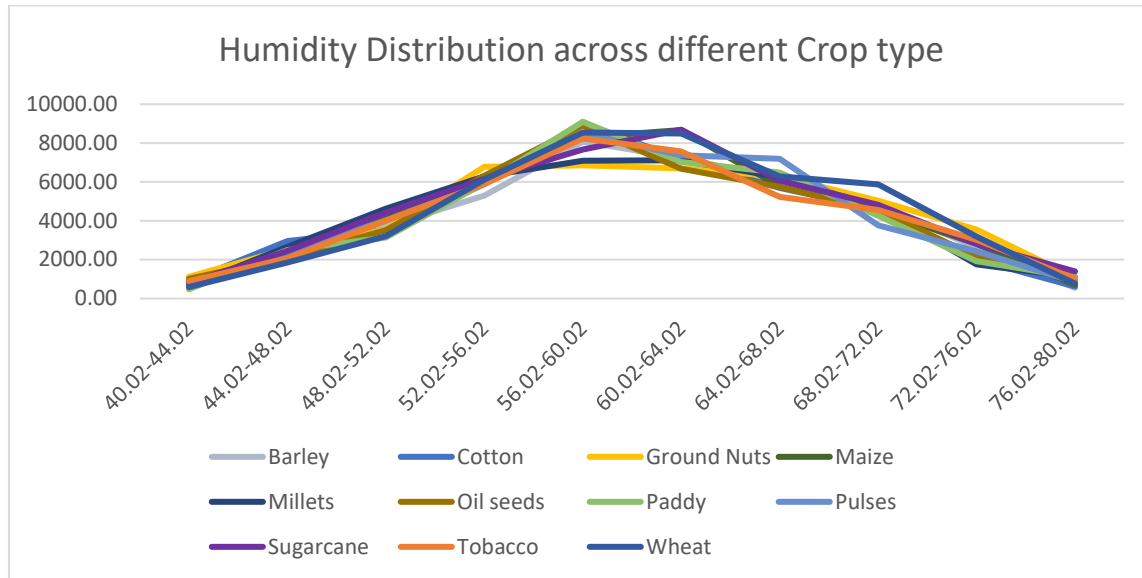
The median aligns closely with the mean, indicating a symmetrical (normal) distribution, which supports the histogram's shape. There are no extreme outliers, suggesting data consistency and homogeneity in moisture levels.

- **Average Humidity:**

The mean humidity of 59.21% aligns well with both the histogram peak and the interquartile range in the box plot.

Together, the histogram, box plot, and average value indicate that humidity levels are tightly clustered around a central range. This consistency points to similar environmental conditions across the samples ; such as being from the same geographic zone or managed with uniform irrigation practices.

Does humidity vary significantly across different crop types?



Row Labels	Average of Humidity
Wheat	60.13
Ground Nuts	59.43
Sugarcane	59.36
Paddy	59.36
Tobacco	59.27
Barley	59.23
Pulses	59.09
Maize	59.01
Oil seeds	58.97
Millet	58.77
Cotton	58.67

- Distribution of humidity across different crop type**

The graph illustrates the distribution of humidity levels across various crop types. Most crops follow a similar pattern, with a noticeable peak in the 56–60% range, and fewer occurrences at the extremes (20–24% and 68–72%).

Paddy and oil seeds are more concentrated within the 56–60% range, indicating that this humidity level might be optimal for their growth. On the other hand, sugarcane and wheat are more prominent in the 60–64% range, suggesting a preference or higher tolerance for more humid

conditions. Notably, wheat also shows higher frequency in the 68–72% range, possibly reflecting its adaptability to high humidity environments.

- **Average Humidity for each crop type**

The average humidity values across crop types are closely clustered, ranging from 58.67% to 60.13%, showing minimal variation. Wheat has the highest average humidity level, though the difference is marginal.

This narrow range indicates that atmospheric conditions or climate management (such as controlled irrigation environments or microclimates) are likely consistent across the observed crop types, contributing to similar humidity levels.

This consistency might indicate that all samples were taken from a single geographic region or a group of regions with similar climate conditions, such as a specific agro-climatic zone. Alternatively, it could suggest the use of standardized irrigation practices across farms, for example, the use of drip or sprinkler systems that not only irrigate but also slightly increase local humidity levels around the crops.

d. Fertilizer Characteristics

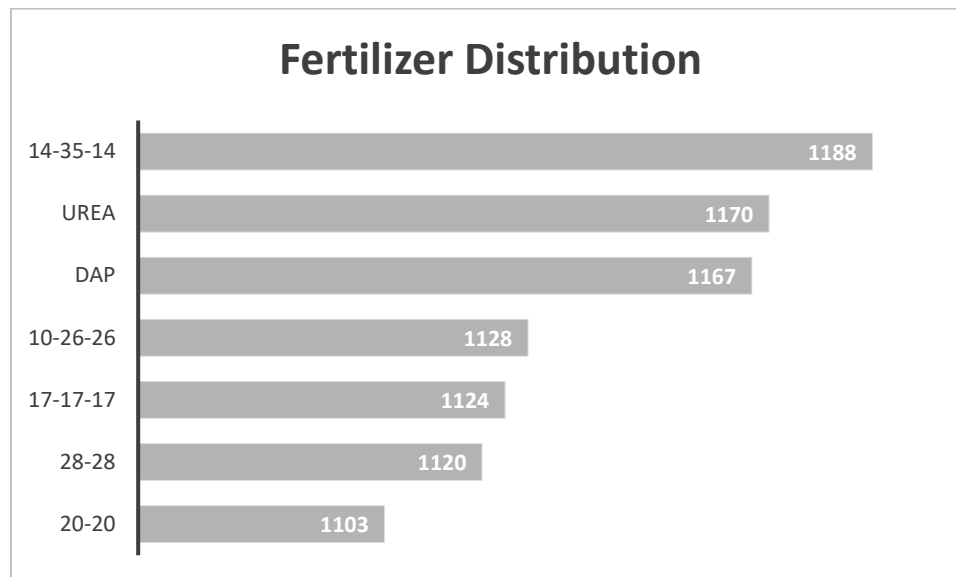
1. Fertilizer name

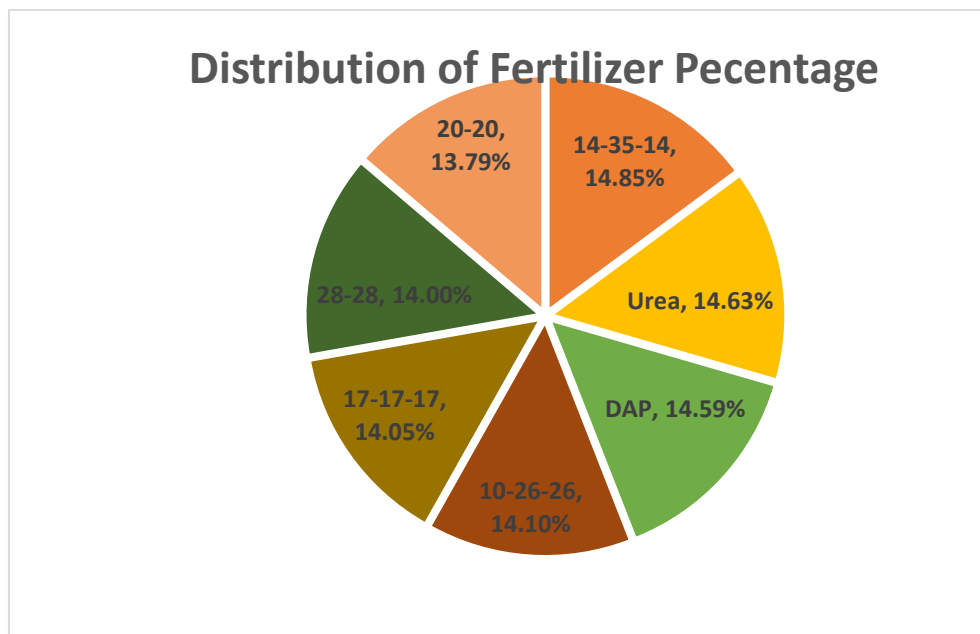
What is the most common soil type reported in the dataset?

Before diving into the analysis, it's important to understand the purpose and application of each fertilizer. The table below summarizes this information.

Fertilizer	N-P-K Composition	Main Use Stage	Why It's Used	Crop/Stage Examples
UREA	46-0-0	Early vegetative growth	Boosts nitrogen for leaf development and greening	Leafy crops (e.g., wheat, maize, rice)
DAP	18-46-0	At sowing/planting	High phosphorus for root growth and early vigor	Legumes, cereals during planting
14-35-14	Moderate N, High P, Moderate K	Early growth phase	Supports root development, stress resistance	Suitable for vegetables, cereals

10-26-26	Low N, High P & K	Flowering & fruiting	Encourages fruit/flower dev, strengthens drought resistance	Fruit crops, vegetables post vegetative stage
17-17-17	Balanced N-P-K	General use throughout the season	Maintains balanced nutrition for all growth phases	Vegetables, cereals, general crops
28-28	High P & K, no N	Flowering & fruiting	Improves yield quality, avoids excess foliage	Fruiting crops, late-stage cereals
20-20	Balanced moderate N-P-K	General maintenance or recovery	Provides essential nutrients without excess	All-purpose fertilizer, crop support





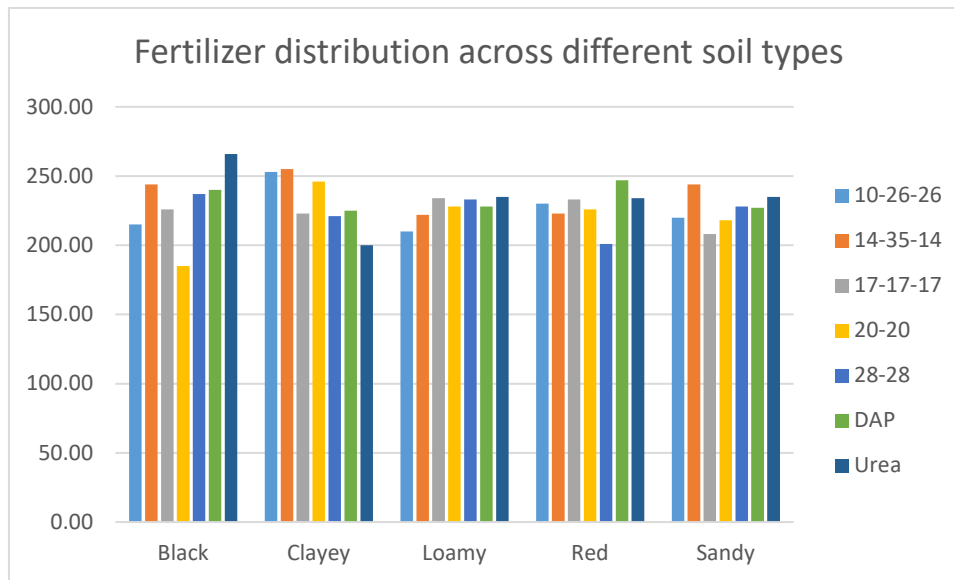
• Fertilizer Distribution Analysis

The data reveals a balanced use of various fertilizers across the fields, with no single type overwhelmingly dominating the landscape:

- 14-35-14 emerges as the most commonly used fertilizer, representing 14.85% of the total usage (1,188 instances).
- UREA follows closely at 14.63% (1,170 instances), reflecting its widespread role as a nitrogen source in crop nutrition.
- DAP (Diammonium Phosphate) is the third most frequently used fertilizer, accounting for 14.59% (1,167 instances), indicating its importance in phosphorus-rich formulations.
- Other fertilizers, such as 10-26-26 (14.10%), 17-17-17 (14.05%), 28-28 (14.00%), and 20-20 (13.79%), show relatively equal distribution, each contributing between 13.7% and 14.1% of total use.

This near-uniform distribution suggests a diverse yet balanced approach to soil fertility management, possibly adapted to the specific nutrient needs of different crops or soil conditions in the region.

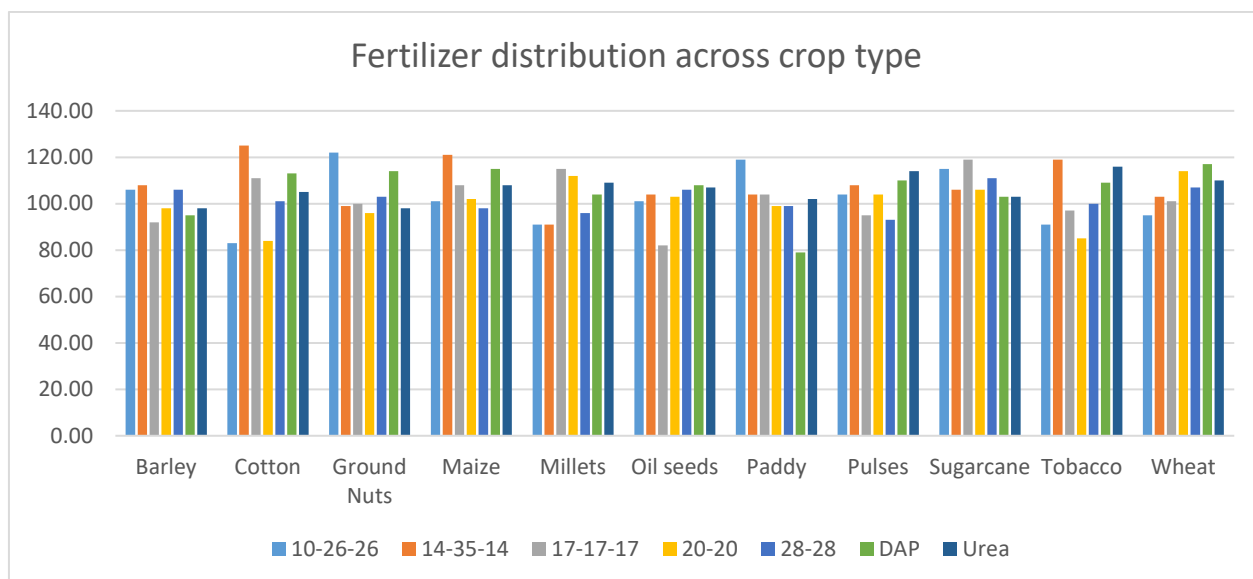
How does fertilizer usage vary across different soil types?



The distribution of fertilizer types varies by soil type, revealing distinct nutrient management strategies:

- **Black Soil:** UREA is the most commonly used, followed by 14-35-14 and DAP. This suggests a nitrogen-focused fertilization approach. 20-20 is used the least, implying a lower preference for balanced blends in this soil type.
- **Clayey Soil:** Farmers favor 14-35-14, followed by 10-26-26 and 20-20, while UREA is the least used. This may indicate a preference for phosphorus- and potassium-rich fertilizers to support root development in dense soils.
- **Loamy Soil:** UREA leads, with 17-17-17 and 28-28 also popular. 10-26-26 is used the least. This balanced usage reflects loamy soil's versatility and responsiveness to varied nutrient mixes.
- **Red Soil:** DAP is the top choice, followed by UREA and 17-17-17. 28-28 is used the least. This may reflect a high phosphorus requirement typical of red soils.
- **Sandy Soil:** 14-35-14 is most preferred, followed by UREA and 28-28, while 17-17-17 ranks lowest. Given sandy soil's quick drainage, the use of phosphorus-rich fertilizers like 14-35-14 may help promote early root growth.

How does fertilizer usage vary across different crop types?



The graph shows the distribution of 8 different fertilizers across 12 crop types. Fertilizer usage varies slightly between crops, but some patterns are clear.

1. Barley

- **Top fertilizers:** 14-35-14, 10-26-26, 28-28
- **Analysis:** Barley requires a good phosphorus supply early in its life cycle (e.g., 14-35-14) to develop a strong root system and early tillering. The balanced NPK in 10-26-26 and potassium-rich 28-28 help ensure healthy spike formation and grain filling.

2. Cotton

- **Top fertilizers:** 14-35-14, DAP, 17-17-17
- **Analysis:** Cotton demands phosphorus for root establishment and flower development, hence the high use of 14-35-14 and DAP. 17-17-17 supports balanced nutrient supply across all growth stages.

3. Ground Nuts

- **Top fertilizers:** 10-26-26, DAP, 28-28
- **Analysis:** Being legumes, groundnuts fix their own nitrogen, so they mainly need phosphorus and potassium. 10-26-26 and DAP enhance root nodulation and early development, while 28-28 supports pod setting and kernel development.

4. Maize

- **Top fertilizers:** 14-35-14, DAP, UREA
- **Analysis:** Maize is a nitrogen-loving crop, requiring UREA for vegetative growth. Phosphorus-rich fertilizers (14-35-14 and DAP) support root development and grain production.

5. Millets

- **Top fertilizers:** 17-17-17, 20-20, UREA
- **Analysis:** Millets are hardy crops but benefit from a balanced NPK (17-17-17) to ensure uniform growth. 20-20 adds extra potassium, improving drought tolerance and grain quality, while UREA provides needed nitrogen.

6. Oil Seeds

- **Top fertilizers:** DAP, UREA, 20-20
- **Analysis:** Oil crops require phosphorus (DAP) for flower and pod formation, while nitrogen (UREA) supports plant growth. 20-20 indicates a balanced input during pod filling to increase oil content.

7. Paddy (Rice)

- **Top fertilizers:** 10-26-26, 14-35-14, 17-17-17
- **Analysis:** Paddy demands steady phosphorus for tillering and root establishment, hence the use of 10-26-26 and 14-35-14. Balanced NPK like 17-17-17 ensures sufficient nutrients throughout grain development. Potassium helps strengthen plants in flooded conditions.

8. Pulses

- **Top fertilizers:** UREA, DAP, 14-35-14
- **Analysis:** Despite nitrogen fixation, pulses use UREA in early stages. DAP and 14-35-14 provide phosphorus needed for root nodulation, flowering, and pod formation.

9. Sugarcane

- **Top fertilizers:** 17-17-17, 10-26-26, 28-28
- **Analysis:** Sugarcane has a long growth cycle and needs sustained nutrients. 17-17-17 ensures balanced growth. 10-26-26 supports early establishment, while 28-28 aids in stalk development and sugar accumulation.

10. Tobacco

- **Top fertilizers:** 14-35-14, UREA, DAP
- **Analysis:** Tobacco needs phosphorus for early root development and nitrogen (UREA) for leaf mass, which is the economic part of the plant. However, excess nitrogen may reduce quality, so doses are controlled. DAP offers a phosphorus-nitrogen mix for early vegetative stages.

11. Wheat

- **Top fertilizers:** DAP, 20-20, UREA

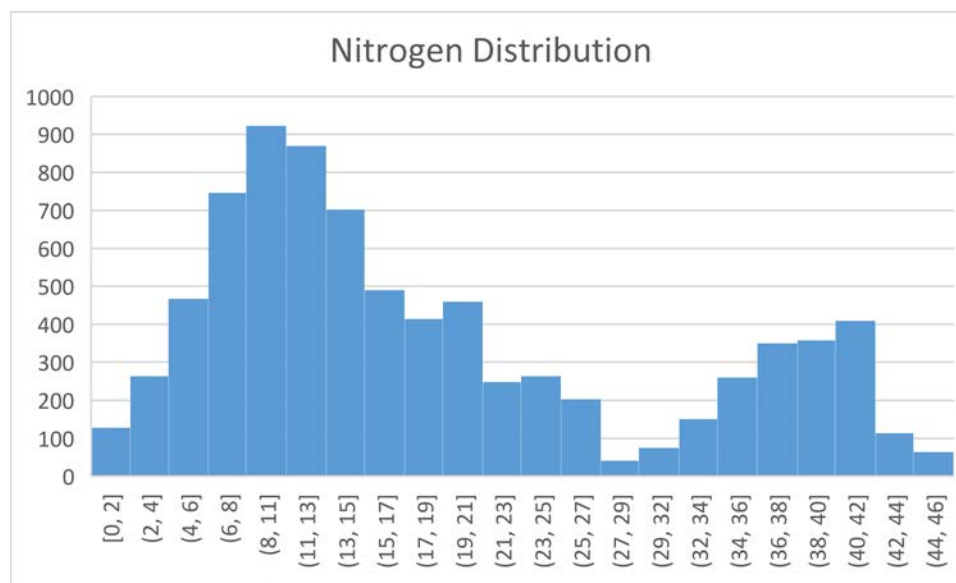
- **Analysis:** Wheat is a heavy nitrogen feeder, with UREA providing key vegetative growth support. DAP supplies early phosphorus, important for tillering. 20-20 ensures balanced nutrient uptake during reproductive stages for better grain quality.

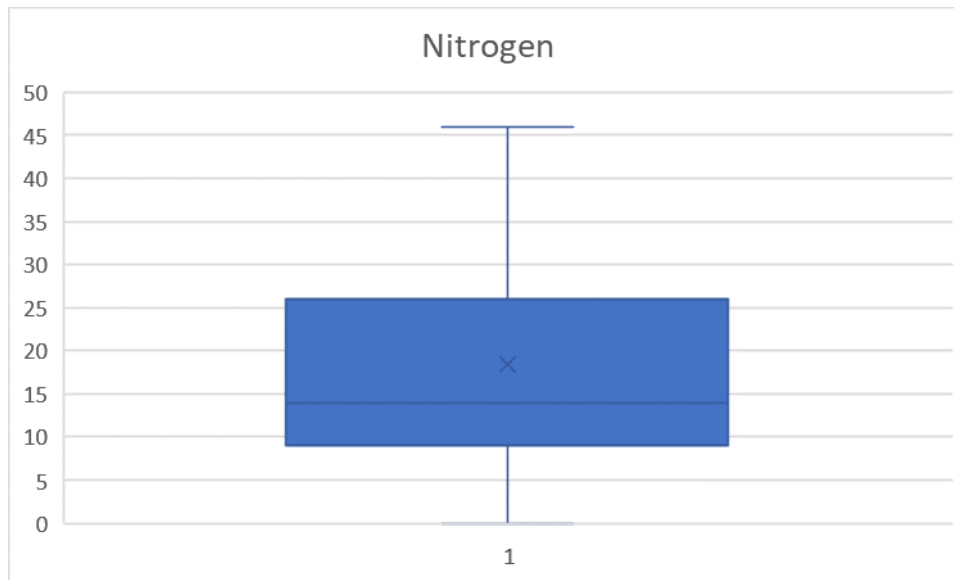
Conclusion:

The fertilizer preferences across crops reveal how nutrient management is tailored to each crop's biological needs. Nitrogen (UREA) is crucial for cereals like maize and wheat due to their leaf and tiller requirements. Phosphorus-rich fertilizers (14-35-14, DAP) are widely used in crops that need strong root systems and flowering, such as cotton and pulses. Balanced fertilizers (like 17-17-17 and 20-20) are common in crops with long growth cycles or multiple harvest stages, like sugarcane and millets. Understanding crop-specific nutrient needs ensures optimal yield and sustainable fertilizer use.

2. Nitrogen

What is the average Nitrogen in the sample?





- **Nitrogen Distribution (Histogram):**

The histogram indicates that the majority of soil types fall within the 8%–13% nitrogen content range, reflecting a preference for moderate nitrogen levels. A secondary concentration is observed between 36% and 42%, likely corresponding to high-nitrogen soils. Very few soil types exhibit extremely low or exceptionally high nitrogen content.

- **Box Plot Interpretation:**

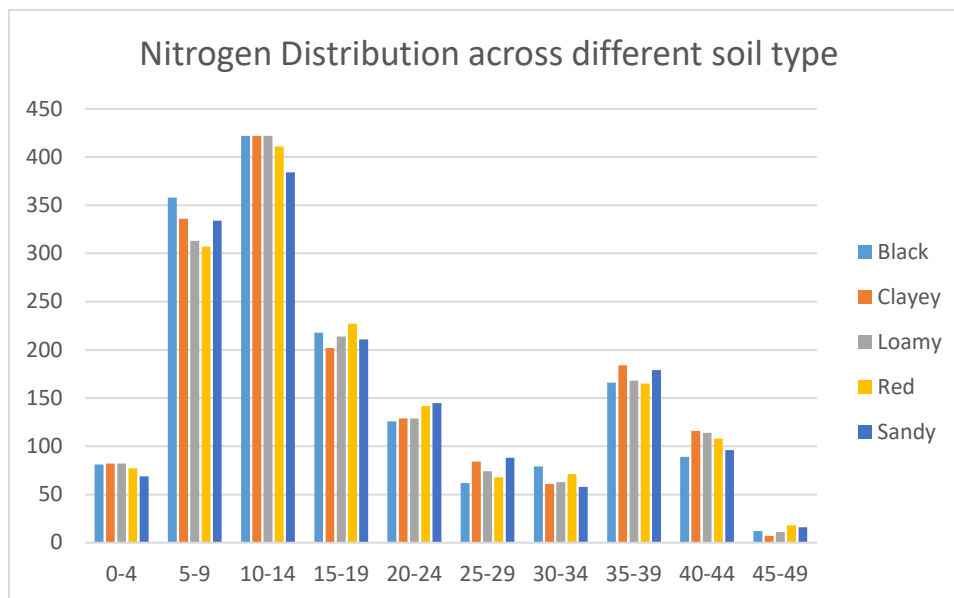
The box plot shows that the nitrogen content in the soil varies widely, with values ranging from close to 0% to approximately 46%. The median nitrogen level is slightly below 15%, while the interquartile range (IQR) extends from around 8% to 26%, indicating that half of the soil samples fall within this moderate range. The mean nitrogen content is slightly higher than the median, suggesting a right-skewed distribution due to the presence of some high-nitrogen values.

- **Average Nitrogen:**

The mean nitrogen content of 18.43% lies above the median, indicating a right-skewed distribution.

This is consistent with the histogram, where most soil samples cluster between 8% and 13%, but a smaller group shows significantly higher nitrogen levels (36%–42%). The box plot also supports this pattern, with a longer upper whisker and a mean higher than the median. These high values, though less frequent, raise the overall average and suggest the presence of nutrient-rich outliers.

Are there significant differences in Nitrogen distributions between Soil Type?



Row Labels	Average of Nitrogen
Red	18.66
Sandy	18.64
Clayey	18.52
Loamy	18.43
Black	17.90

- **Distribution of nitrogen across different soil type**

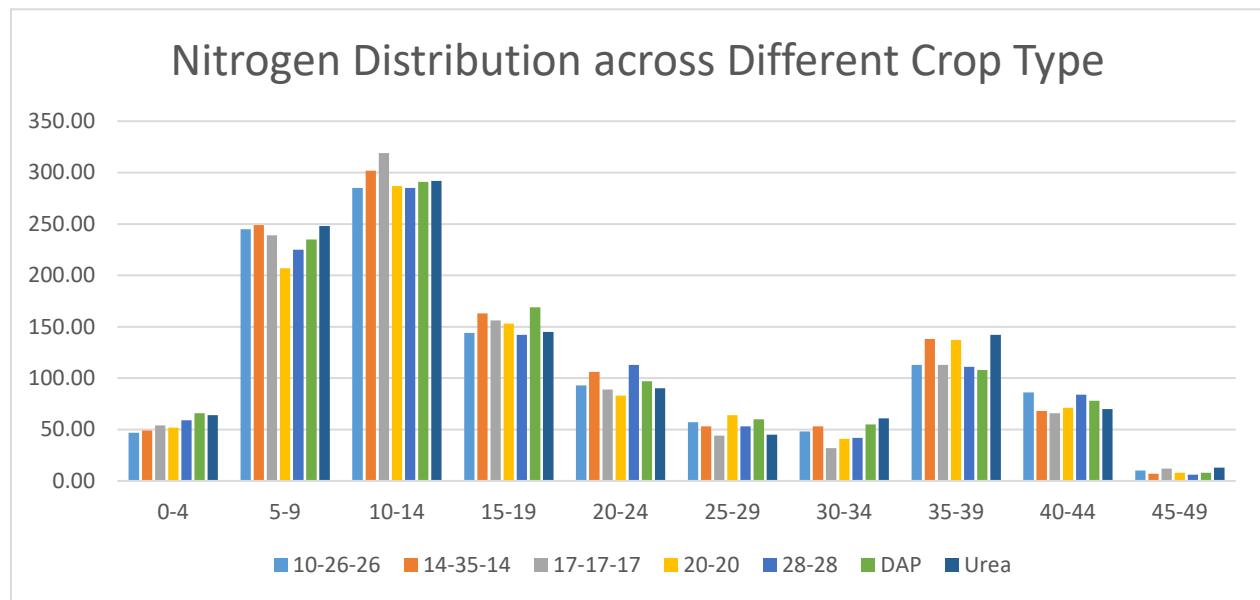
The graph illustrates the distribution of nitrogen across various soil types. The bimodal nitrogen distribution observed across soil types likely reflects differences in soil composition, fertilization practices, and land management. Black, Loamy, and Clayey soils cluster around the 10–14% nitrogen range due to their moderate organic matter content, good water retention, and balanced microbial activity, which naturally support steady nitrogen levels. In contrast, the 35–39% nitrogen peak seen in Clayey and Sandy soils may result from intensive nitrogen fertilization—either to overcome Clayey’s nutrient-binding nature or sandy soil's high leaching potential. Additionally, regional crop demands and historical management practices could further influence these nitrogen concentrations.

- **Average Nitrogen for each crop type**

The average nitrogen content across different soil types is tightly clustered, ranging from 17.90% to 18.66%, indicating minimal variation. Although Red soil exhibits the highest average, the difference is relatively small.

This narrow range suggests that fertilization practices are generally consistent across the soil types studied, leading to comparable nitrogen levels.

Are there significant differences in Nitrogen distributions between Crop Types?



Row Labels	Average of Nitrogen
Millet	19.17
Maize	18.90
Sugarcane	18.76
Tobacco	18.61
Paddy	18.52
Oil seeds	18.41
Ground Nuts	18.31
Pulses	18.09
Wheat	18.06
Cotton	17.98
Barley	17.87

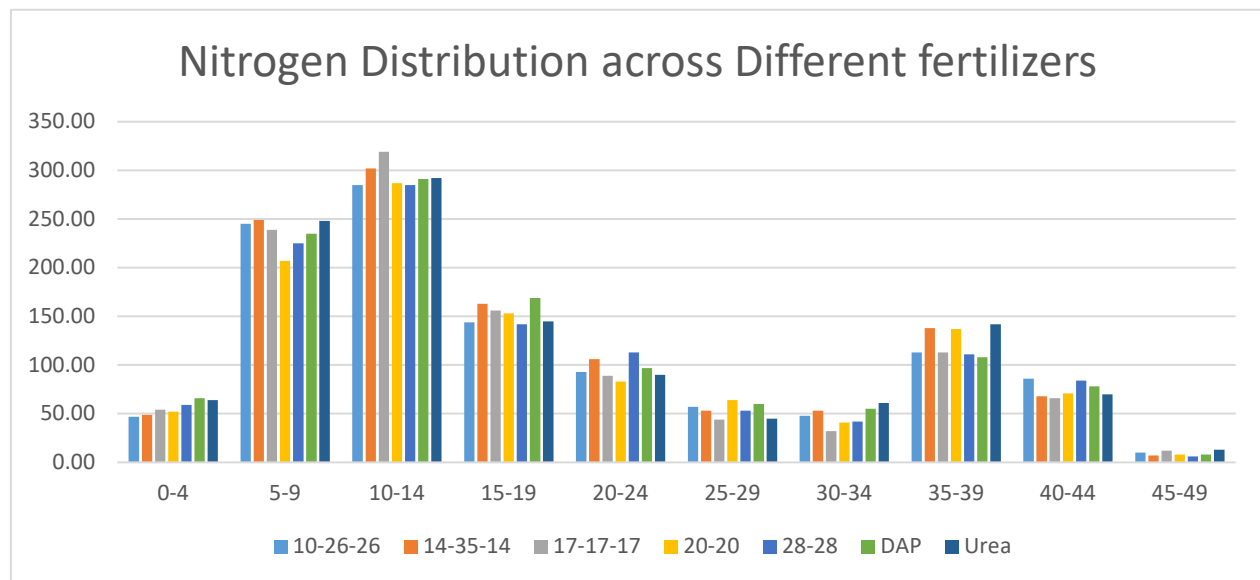
- Distribution of nitrogen across different crop type**

The graph shows the distribution of nitrogen levels among various crop types. Most crops exhibit a similar trend, with two noticeable peaks, one between 10–14% and another between 35–39%. Barley and sugarcane are primarily concentrated within the 10–14% range, suggesting that moderate nitrogen levels may be optimal for their growth. In contrast, maize shows a stronger presence in the 35–39% range, which may indicate nitrogen fixation or a higher nitrogen demand for optimal development.

- Average nitrogen for each crop type**

The graph and average nitrogen values indicate a consistent pattern in fertilization practices or soil nitrogen availability across different crop types, ensuring an adequate yet balanced nutrient supply. While most crops appear to thrive within a moderate nitrogen range, certain crops like maize may require or benefit from higher nitrogen levels under specific conditions.

Are there significant differences in Nitrogen distributions between Fertilizers?



Row Labels	Average of Nitrogen
20-20	18.85
10-26-26	18.63
Urea	18.60
28-28	18.50
14-35-14	18.48
DAP	18.19
17-17-17	17.76

- Distribution of nitrogen across different fertilizers**

The graph shows the distribution of nitrogen levels among various fertilizer, Overall, most fertilizers follow a similar trend, with higher concentrations observed in the nitrogen range of 5–14%. Fertilizers such as 17-17-17, 14-35-14, and 28-28 are the most commonly used within this range. Another noticeable peak occurs between 35–44%, where UREA, 20-20, and 14-35-14 show the greatest presence.

- Average Nitrogen for each fertilizer**

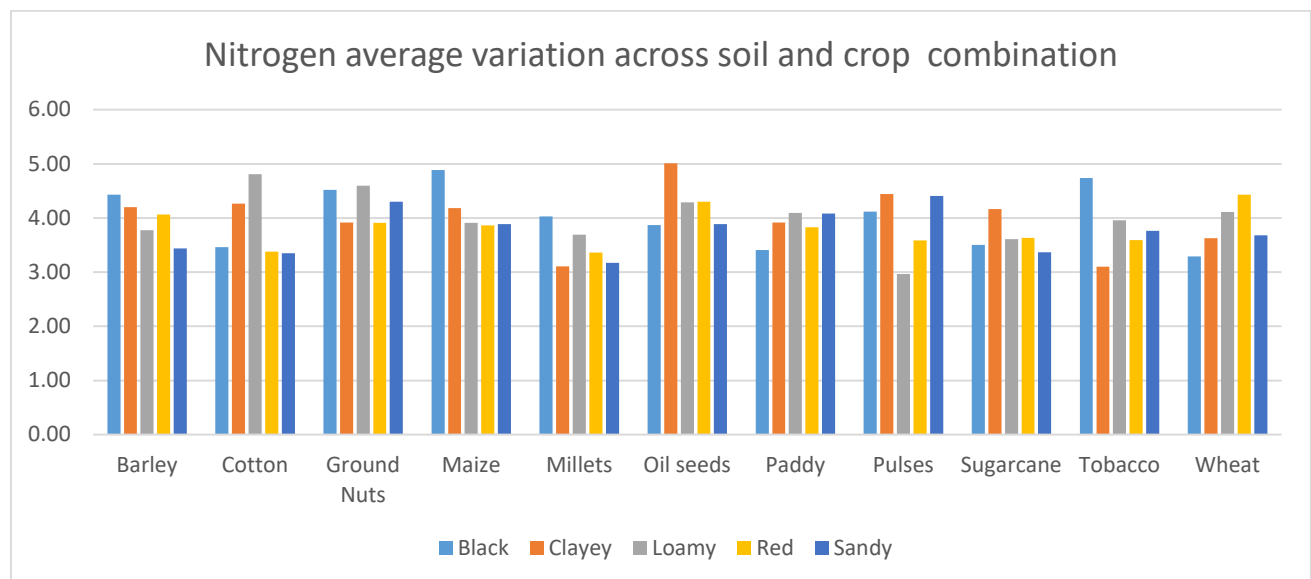
The average nitrogen values across all fertilizers remain relatively close, ranging from 17.76% to 18.85%, with 20-20 displaying the highest average.

The combination for graph and average highlights that most fertilizers are applied to maintain moderate nitrogen levels in the soil—specifically between 5–14%. This aligns with standard crop nutrition strategies, where moderate nitrogen promotes healthy vegetative growth without causing nutrient imbalances or excessive foliage at the expense of yields.

The second peak between 35–44%, where UREA and 20-20 are prominent, may indicate targeted fertilization for high-demand crops. In the field, such spikes can occur during intensive cultivation cycles, especially for crops like maize or wheat, which are nitrogen-hungry.

How do different combinations of soil and crop types influence nitrogen levels?

Average of Nitrogen Row Labels	Column Labels				
	Black	Clayey	Loamy	Red	Sandy
Millets	18.88	20.92	17.28	20.51	18.38
Paddy	17.79	18.97	18.50	19.20	17.97
Maize	17.47	18.77	20.15	18.42	19.65
Ground Nuts	17.38	18.72	18.18	18.28	18.95
Oil seeds	17.98	18.62	18.77	17.98	18.78
Wheat	17.48	18.61	19.64	17.24	17.39
Tobacco	18.33	18.41	19.52	17.61	19.11
Pulses	18.50	18.41	17.85	18.23	17.26
Cotton	18.41	17.57	16.24	19.81	18.09
Sugarcane	17.43	17.52	18.89	19.91	20.14
Barley	16.98	17.22	17.81	18.16	18.97



- **average nitrogen levels across different crop and soil type combinations**

Millets exhibit the highest nitrogen levels in clayey and red soils, as does paddy. Maize shows elevated nitrogen in loamy and sandy soils, while groundnuts peak in clayey and sandy soils. Oil seeds show higher nitrogen in sandy and loamy soils, and wheat in loamy and clayey. Tobacco performs better in loamy and sandy soils, while pulses show higher nitrogen in black and clayey soils. Cotton records its highest nitrogen in red and black soils. Sugarcane peaks in sandy and red soils, and barley in sandy and red as well.

There is a clear interaction between crop type and soil type in terms of nitrogen levels. Different crops tend to show higher nitrogen concentrations in specific soil types, indicating either:

1. **Soil preference of the crops**, where certain crops naturally thrive better in soils that retain more nitrogen.
2. **Targeted fertilization practices**, where farmers may be adjusting nitrogen input based on the crop-soil combination.

By examining the heatmap alongside the graph, we can observe that:

Clayey and red soils appear to support higher nitrogen levels for crops like millets, paddy, and cotton.

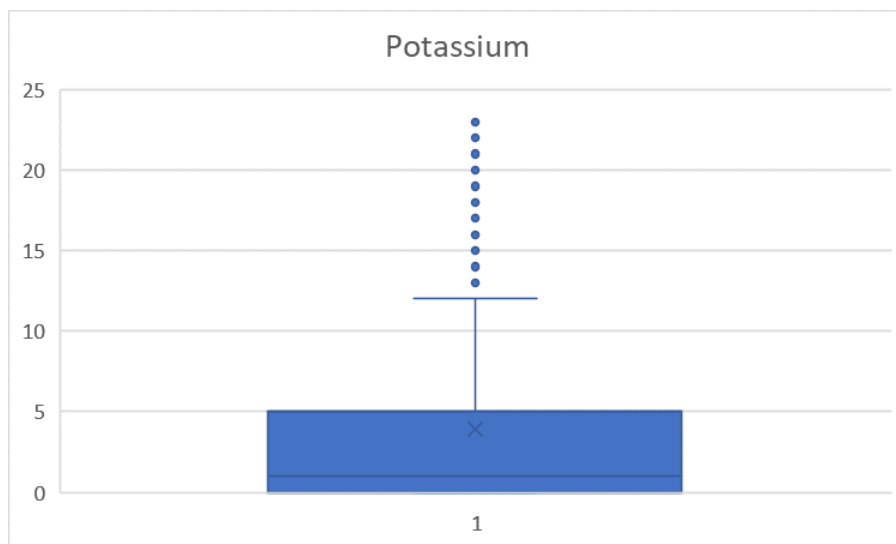
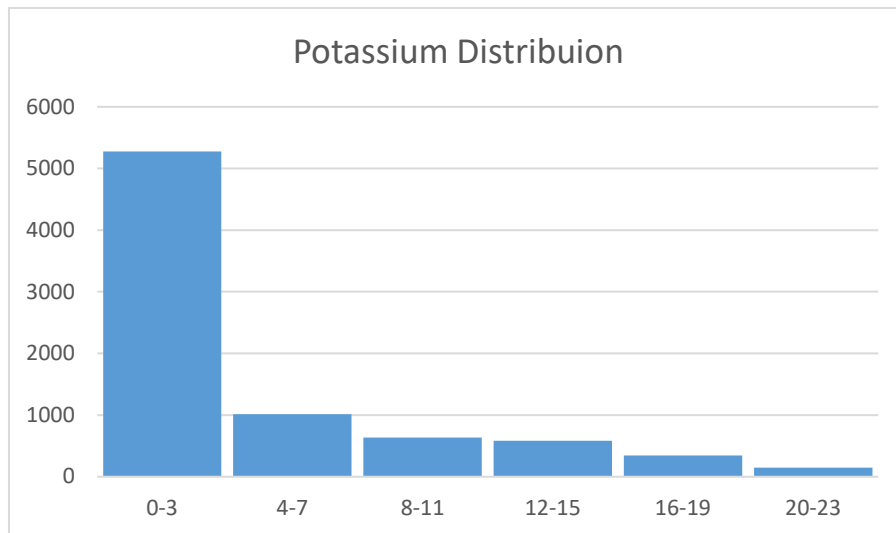
Loamy soils are favorable for crops like maize, tobacco, and wheat, while sandy soils are more suited for barley, oil seeds, and sugarcane.

Black soil shows higher nitrogen for pulses and cotton, suggesting either good nitrogen retention or fertilization.

Overall, this suggests that nitrogen management is likely crop- and soil-specific, and optimizing this relationship can lead to more efficient fertilizer use and better crop performance.

3. Potassium

What is the average Potassium in the sample?



- **Potassium Distribution:**

The bar chart shows that the majority of soil samples have potassium levels in the 0–3 range, with over 5,000 occurrences; indicating a very high concentration of low potassium levels. The frequency drops significantly in the subsequent ranges (4–7, 8–11, etc.), showing that only a small fraction of samples contains medium to high levels of potassium. This suggests that potassium deficiency may be a widespread issue in the dataset.

- **Box Plot Interpretation**

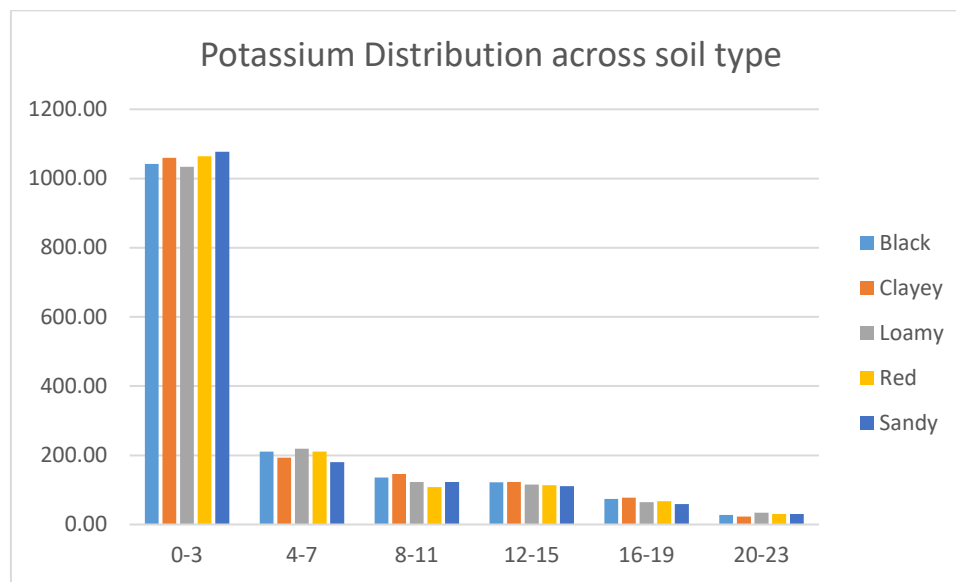
The box plot illustrates that potassium levels in the data are highly skewed towards the lower end. The median value lies near the lower quartile, confirming that most values are clustered around low potassium concentrations. There are many outliers above the upper whisker, indicating the presence of some samples with unusually high potassium levels, but these are rare compared to the rest.

- **Average Potassium**

The mean potassium content of 3.91% gives us an overall average level of potassium in the soil samples. However, if this value is paired with a boxplot, and we know that the median is lower than 3.91%, we can conclude:

- The data is right-skewed, meaning a few samples have very high potassium values.
- Most of the soil samples have potassium below 3.91%.
- These higher values are pulling up the mean, even though they are not common.

Are there significant differences in Potassium distributions between Soil Type?



Row Labels	Average of Potassium
Clayey	4.01
Black	4.01
Loamy	3.99
Red	3.82
Sandy	3.75

• Distribution of Potassium across different soil type

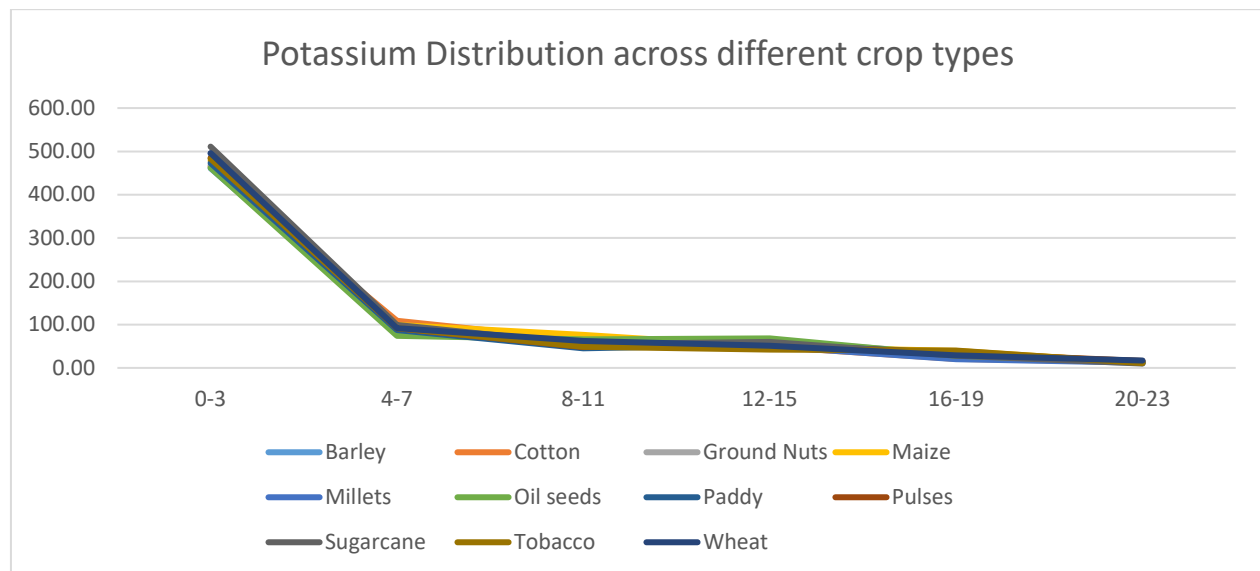
This graph illustrates the distribution of potassium levels across different soil types. The x-axis represents potassium percentage ranges (bins), while the y-axis indicates the frequency of soil samples within each range. Each soil type is represented by a different color. All five soil types follow a similar pattern, with a noticeable peak in the 0–3% potassium range. This suggests that the majority of soils, regardless of type, have relatively low potassium content.

• Average Potassium for each crop type

All soil types exhibit low average potassium levels, ranging between 3.75% and 4.01%, with slightly higher concentrations observed in clayey and black soils.

Since potassium is essential for crop growth, supporting root development, disease resistance, and water regulation, the low potassium content suggests a need for potassium supplementation across most soils. Clayey and black soils might require slightly less amendment compared to others, but overall, regular monitoring and targeted fertilization will be crucial for maintaining soil health and optimizing crop yield.

Are there significant differences in Potassium distributions between Crop Type?



Row Labels	Average of Potassium
Oil seeds	4.27
Ground Nuts	4.25
Maize	4.14
Barley	3.96
Pulses	3.93
Cotton	3.88
Paddy	3.86
Tobacco	3.84
Wheat	3.83
Sugarcane	3.66
Millets	3.48

- Distribution of Potassium across different crop type

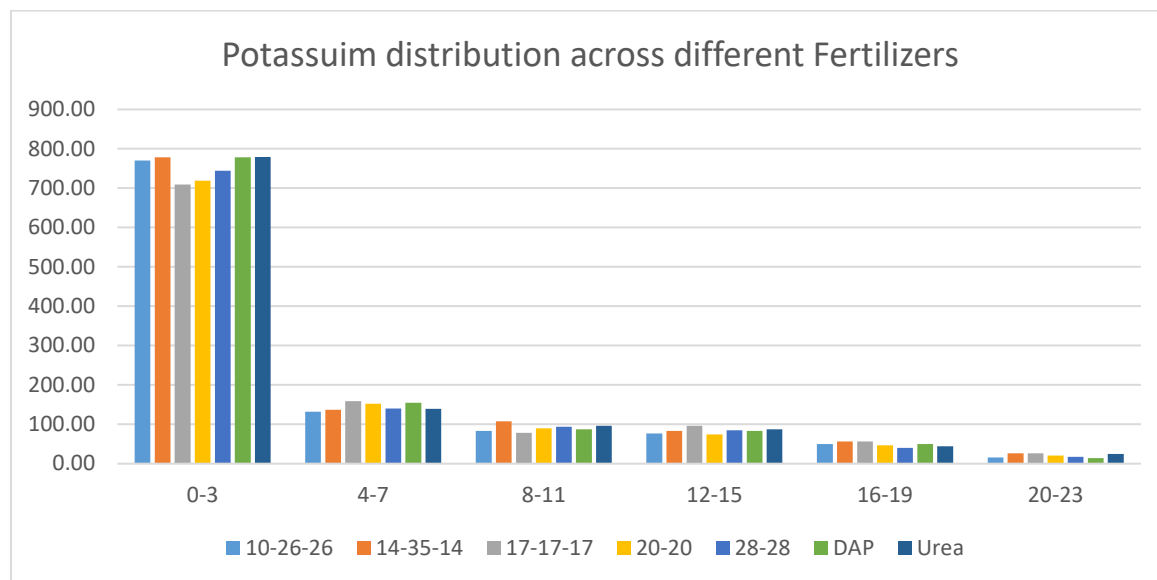
The graph illustrates the distribution of potassium levels across various crop types. A common trend is observed among all crops, with one prominent peak, one in the 0–3% range, indicating a high concentration of crops with low potassium levels. Notably, groundnut shows a slight increase in frequency within this range, suggesting a particular tendency toward lower potassium content.

- **Average Potassium for each crop type**

The average potassium content for different crops ranges narrowly between 3.48% and 4.27%. This small variation indicates that, overall, the crops in the dataset require or contain similar levels of potassium.

The consistency in potassium distribution and average content across crops suggests a generally low potassium presence in the dataset. This could reflect either the natural soil potassium availability or the nutrient uptake pattern of the crops. From a practical standpoint, it highlights the importance of potassium fertilization strategies across most crop types to ensure balanced nutrition and optimal yield, especially for crops like groundnut which show slightly higher sensitivity to lower potassium levels.

Are there significant differences in Potassium distributions between Fertilizers?



Row Labels	Average of Potassium
17-17-17	4.27
14-35-14	4.08
Urea	3.91
20-20	3.91
28-28	3.81
DAP	3.73

- **Distribution of Potassium across different fertilizers**

The graph displays the distribution of potassium levels across different fertilizers use. Overall, most fertilizers follow a similar trend, with a high concentration observed in the 0–3% potassium range. Fertilizers such as UREA, DAP, and 14-35-14 are particularly prevalent within this range, indicating their frequent use in scenarios where lower potassium levels are sufficient or expected.

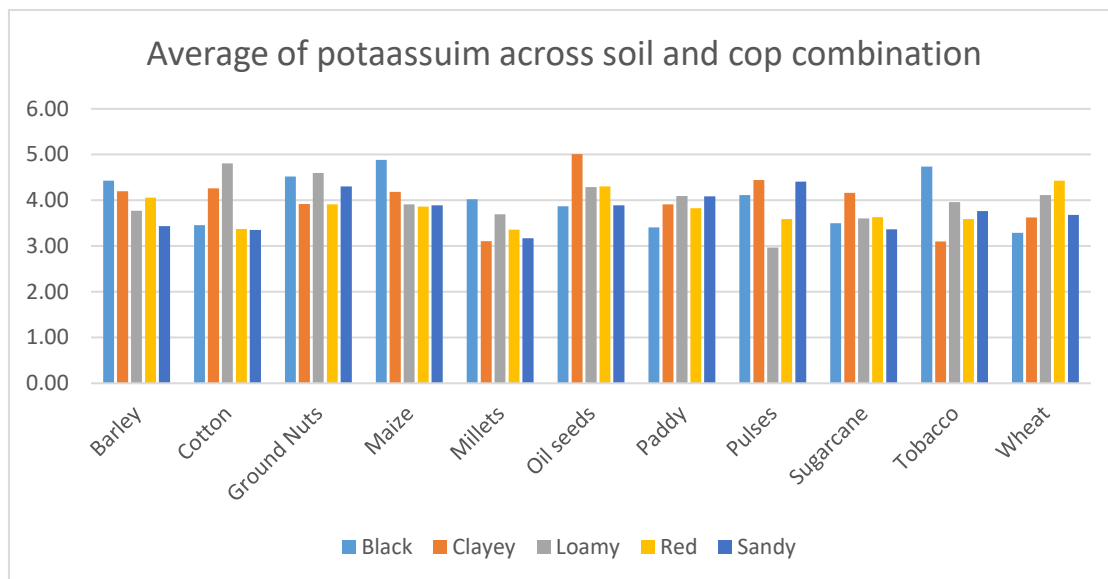
- **Average Potassium for each fertilizer**

The average potassium values across all fertilizer types fall within a relatively narrow range; from 3.70% to 4.27%. Among them, 17-17-17 stands out with the highest average potassium content, suggesting a more balanced NPK formulation compared to the others.

This analysis suggests that most fertilizers are applied in soils with low potassium availability, either because of naturally deficient soils or because those crops do not require high potassium input. The relatively uniform average across fertilizers also implies that fertilizer selection may be driven more by crop needs or nutrient balance rather than correcting severe potassium shortages.

How do different combinations of soil and crop types influence Potassium levels?

Average of Potassium Row Labels	Column Labels				
	Black	Clayey	Loamy	Red	Sandy
Barley	4.43	4.20	3.77	4.06	3.44
Cotton	3.46	4.26	4.81	3.38	3.35
Ground Nuts	4.52	3.92	4.60	3.91	4.30
Maize	4.88	4.18	3.91	3.86	3.89
Millets	4.03	3.11	3.69	3.36	3.17
Oil seeds	3.87	5.01	4.29	4.30	3.89
Paddy	3.41	3.91	4.10	3.83	4.08
Pulses	4.12	4.44	2.96	3.59	4.41
Sugarcane	3.50	4.17	3.61	3.63	3.36
Tobacco	4.74	3.10	3.96	3.59	3.77
Wheat	3.29	3.63	4.11	4.43	3.68



- **average Potassium levels across different crop and soil type combinations**

Maize reaches its **highest average potassium** in **black soil (4.88)**, followed by **clayey (4.18)**.

Oil Seeds show a strong preference for **clayey soil (5.01)** — the **highest potassium level overall** across all combinations.

Cotton performs best in **loamy soil (4.81)**, hinting at nutrient compatibility in that texture.

Pulses have their **lowest potassium average (2.96)** in **loamy soil**, showing an exception to the general trend of loamy being favorable.

Tobacco reaches **4.74** in **black soil**, suggesting better potassium retention or uptake in that type.

Soil Management: Farmers can optimize potassium fertilization by aligning **soil types** with **specific crop needs**. For instance, if planting oil seeds, using clayey soil might reduce the need for potassium supplementation.

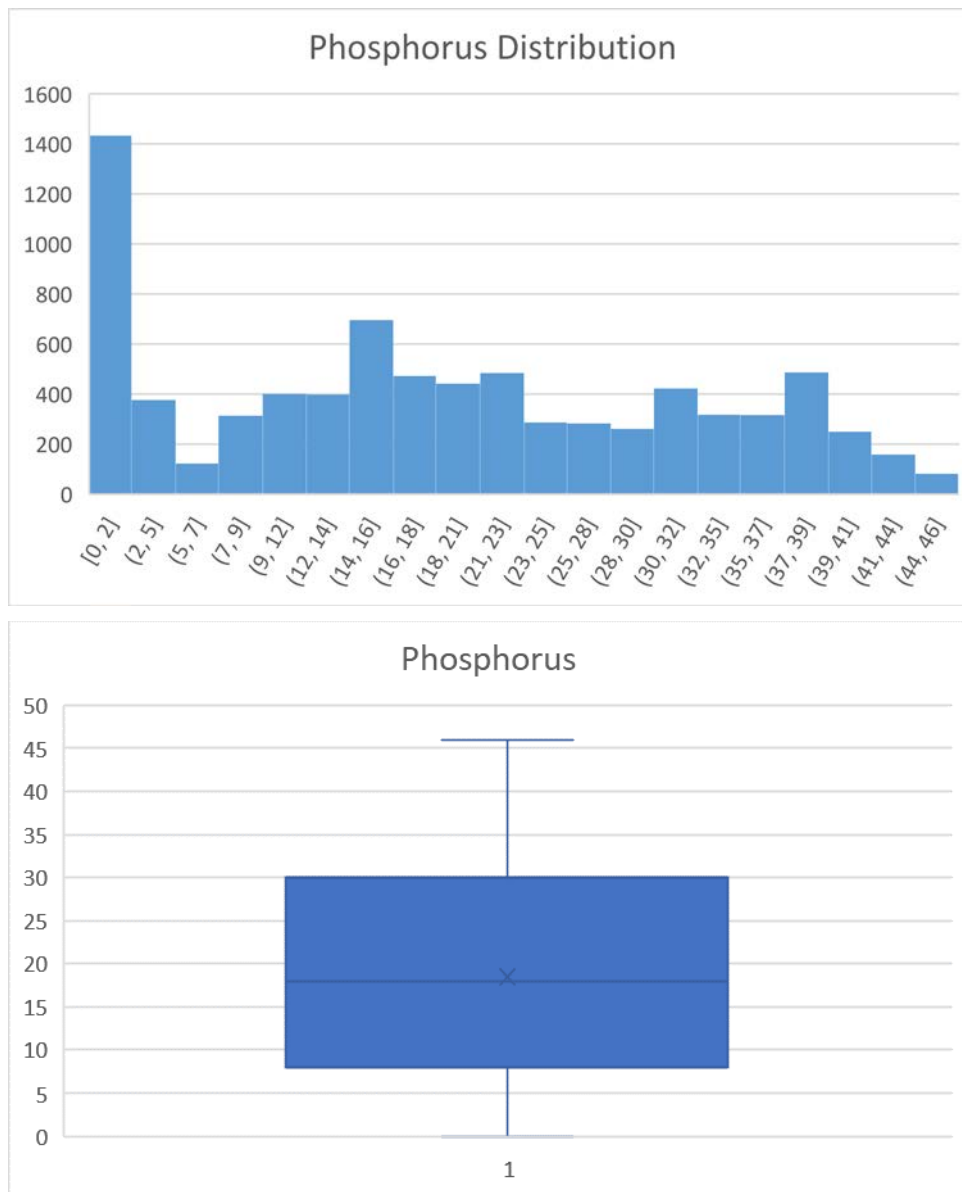
Fertilizer Efficiency: Understanding natural potassium content helps **avoid over-fertilization**, saving costs and reducing environmental impact.

Crop Planning: Rotating crops that prefer high-potassium soils with those that perform well under lower potassium conditions can improve **long-term soil health** and productivity.

Potassium availability varies depending on the soil-crop interaction, not just one or the other. While clayey and black soils generally support higher potassium levels, exceptions exist. To improve yield and efficiency, farmers should consider both soil type and crop nutrient demand when planning agricultural activities.

4. Phosphorus

What is the average Phosphorus in the sample?



- **Phosphorus Distribution**

The bar chart reveals that the majority of soil samples have phosphorus levels in the 0–2% range, with over 1,400 occurrences, indicating a significant concentration of low phosphorus content. Additional peaks are observed in the 14–16% range with approximately 700 samples, and in the 37–39% range with around 500 samples. In contrast, the frequency of samples in others ranges drops below 500, suggesting that medium to high phosphorus levels are relatively uncommon. Overall, the distribution implies that phosphorus deficiency is prevalent across the dataset, which may require targeted soil management or fertilization interventions.

- **Box Plot Interpretation**

The box plot illustrates the distribution of phosphorus levels across the dataset. The interquartile range (IQR); which represents the middle 50% of the data—extends from approximately 7% to 30%, indicating that most soil samples have phosphorus levels within this range. The median is around 18%, showing the central tendency of the data.

The mean lies slightly above the median, suggesting a slight right skew in the distribution—meaning some higher phosphorus values are pulling the average upward. The whiskers extend from close to 0% up to around 46%, showing a relatively wide spread of values with no extreme outliers, which suggests that the dataset contains both low and high phosphorus levels but without extreme anomalies.

- **Average Phosphorus**

The mean phosphorus content of 18.52% represents the average phosphorus level across all soil samples. However, given the distribution shown in histogram and box plot, where:

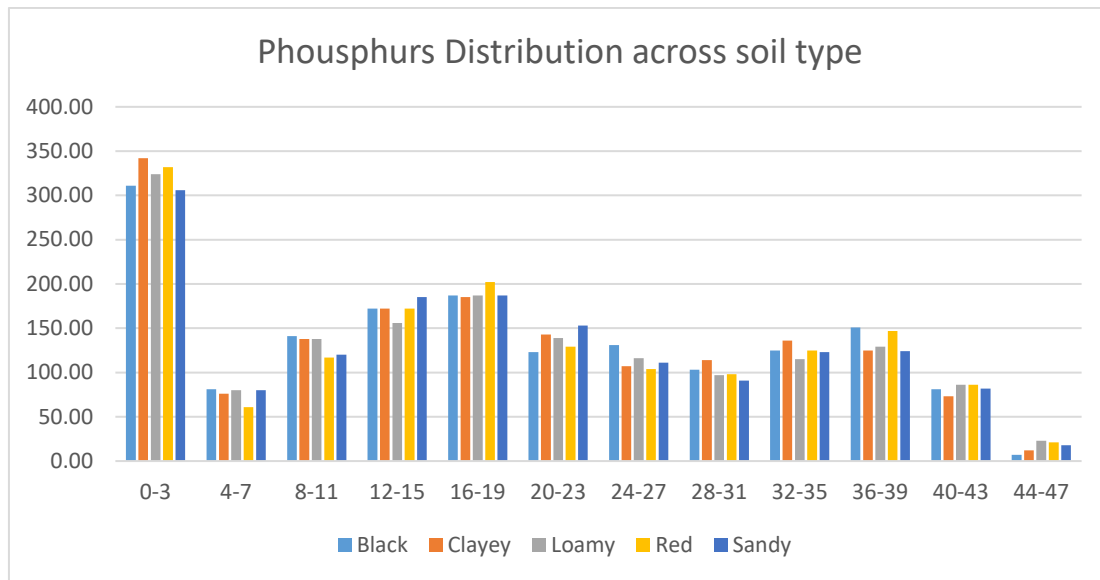
- Most samples are concentrated in the 0–2% and 14–16% ranges,
- The box plot shows the median is below the mean,
- And there's a right-skewed distribution (with fewer high values),

The mean is being pulled upward by a smaller number of samples with very high phosphorus levels (e.g., in the 37–39% range or beyond).

This means does not reflect the most common phosphorus level in this dataset, it's higher than what most soils actually contain.

Using the mean alone could mislead decision-making if you assume most soils have around 18.5% phosphorus. In reality, many soils fall well below this value, particularly in the deficient 0–2% range. For effective soil and crop management, it's more useful to combine the mean with the median, histogram, and box plot insights to make data-driven fertilization decisions.

Are there significant differences in Phosphorus distributions between Soil Type?



Row Labels	Average of Phosphorous
Black	18.66
Clayey	18.09
Loamy	18.50
Red	18.82
Sandy	18.49

- Distribution of Phosphorus across different soil type**

This chart shows how phosphorus levels are distributed across five soil types: Black, Clayey, Loamy, Red, and Sandy.

All soil types show the highest concentration in the 0–3% range, with clayey soil having the highest count, followed closely by red and loamy. This suggests that phosphorus deficiency is widespread across all soils.

The 16–19% range also shows a noticeable cluster across all soil types, particularly for red soil, suggesting a second concentration point of moderately phosphorus-rich soils.

Another visible grouping appears between 36–39%, again across all soil types, with black and red soils slightly more represented.

The frequency generally declines in very high phosphorus ranges (40–47%), especially for loamy and red soils, where values become minimal.

- Average Phosphorus for each crop type**

The table shows that the average phosphorus content is relatively consistent across all soil types, ranging narrowly from 18.09% to 18.82%.

Red soil has the highest average phosphorus level at 18.82%, closely followed by black (18.66%) and loamy (18.50%).

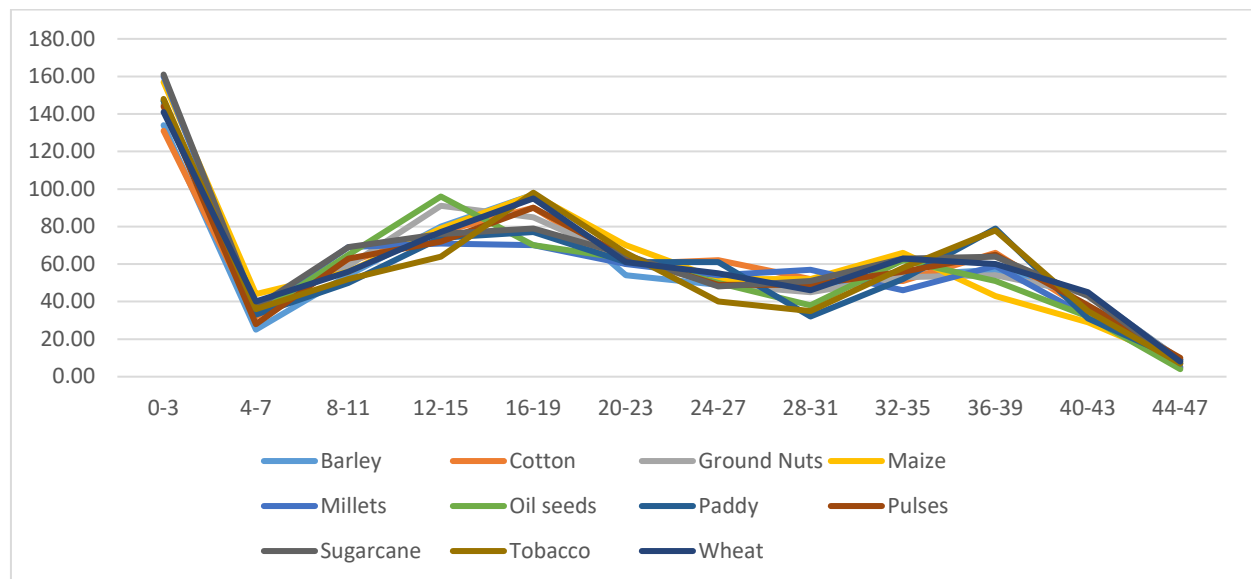
Clayey soil has the lowest average at 18.09%, though the difference remains small.

Despite the small differences in averages, the previous distribution chart shows that all soil types had a strong peak at the lowest phosphorus range (0–3%). This suggests:

While the average values appear balanced, they are influenced by a minority of high-phosphorus samples (e.g., in the 36–39% or 40+ ranges), which skew the mean upward.

In reality, a large portion of each soil type still falls in the deficient zone, as seen in the histogram.

Are there significant differences in Phosphorus distributions between Crop Type?



Row Labels	Average of Phosphorous
Barley	19.03
Cotton	19.00
Wheat	18.97
Pulses	18.91
Paddy	18.75
Tobacco	18.71
Ground Nuts	18.47

Sugarcane	18.47
Millet	17.83
Maize	17.78
Oil seeds	17.75

- **Distribution of Phosphorus across different crop type**

All crop types show a sharp peak in the 0–3% phosphorus range, with values ranging from 130 to over 160 samples.

This confirms that phosphorus deficiency is common across nearly all crops.

After the 0–3% bin, there's a dip in the 4–7% range, then a rise again in the 12–19% range, indicating a secondary concentration of phosphorus in moderate levels.

Oil seeds and ground nut show relatively higher phosphorus values in the 12–19% range, possibly indicating a stronger phosphorus demand or better soil fertility in those crops.

Tobacco has noticeable peaks in the 16–19% and 32–35% ranges, suggesting more samples with mid-to-high phosphorus content.

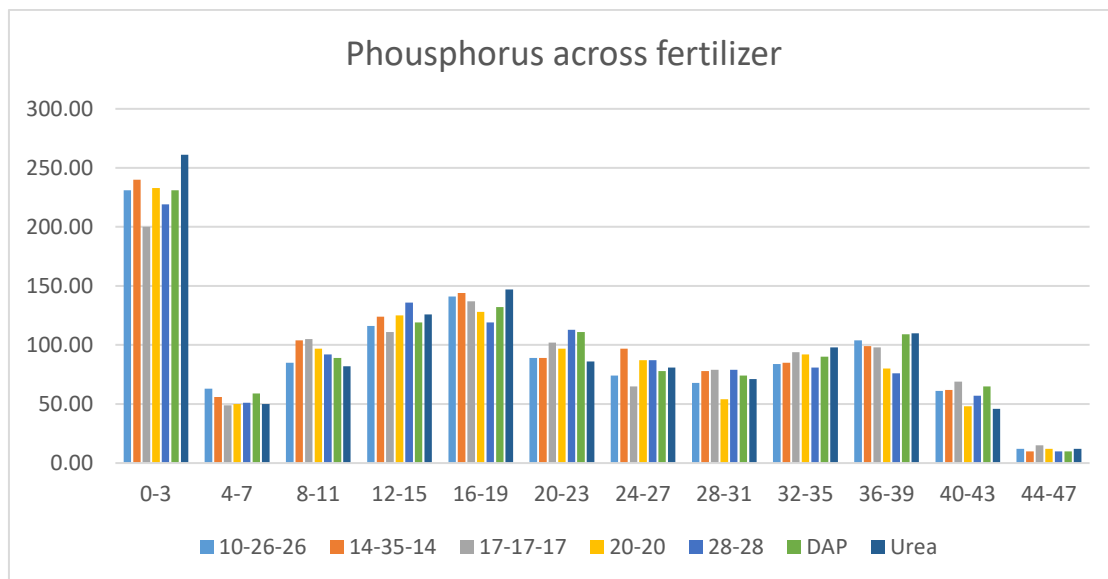
From 20% onward, most crops show a gradual decline in frequency as phosphorus levels increase, very few samples are found in the 44–47% range.

- **Average Phosphorus for each crop type**

The average phosphorus levels across crop types are relatively close, ranging from 17.75% (Oil seeds) to 19.03% (Barley).

Although the average phosphorus values across crops are quite similar, subtle differences highlight crop-specific nutrient patterns. Crops like barley and cotton may require or retain slightly more phosphorus, while oil seeds and maize may tolerate lower levels. These insights support the need for crop-specific phosphorus management, especially when planning fertilization strategies or assessing soil fertility across diverse fields.

Are there significant differences in Phosphorus distributions between Fertilizers?



Row Labels	Average of Phosphorous
10-26-26	18.47
14-35-14	18.38
17-17-17	19.37
20-20	17.97
28-28	18.36
DAP	18.85
Urea	18.17

- **Distribution of Phosphorus across different crop type**

This chart shows how phosphorus levels are distributed across different fertilizer types, including 10-26-26, 14-35-14, 17-17-17, 20-20, 28-28, DAP, and UREA.

Highest peak at 0–3%:

All fertilizers show a strong concentration in the 0–3% range, with Urea and 10-26-26 leading in frequency.

- **Average Phosphorus for each fertilizer**

The table shows that the average phosphorus levels across fertilizers range from 17.97% to 19.37%, indicating a relatively narrow spread. However, there are some notable distinctions:

17-17-17 has the highest average phosphorus (19.37%), which aligns with its balanced NPK composition.

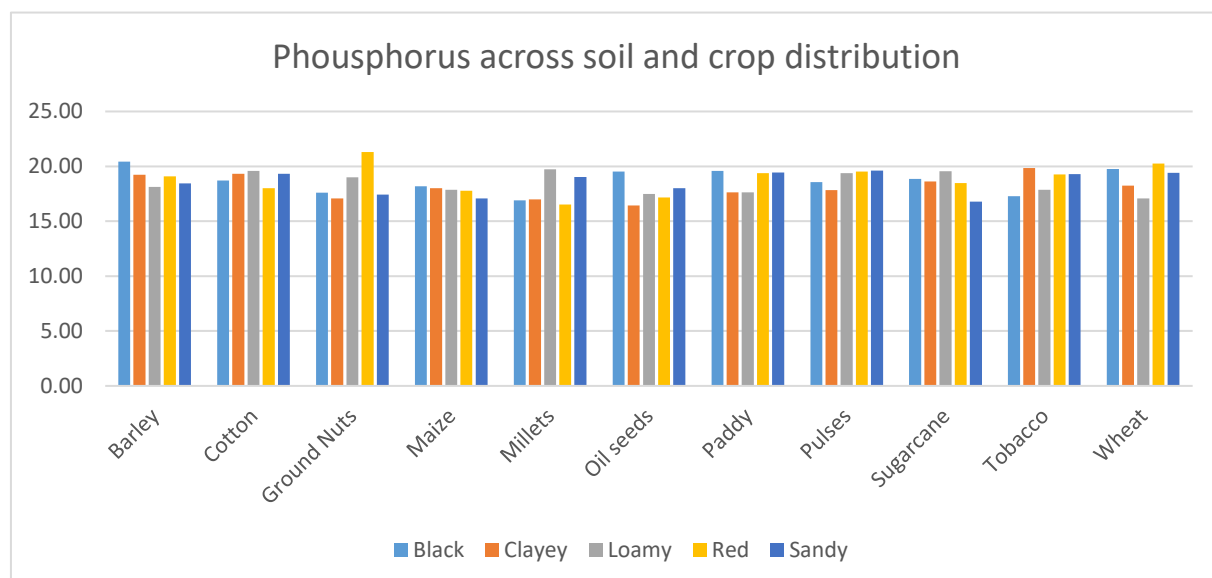
DAP (18.85%) also shows a high phosphorus average, reflecting its phosphorus-rich formula and targeted use in deficient soils.

20-20 and Urea have the lowest averages (17.97% and 18.17% respectively). This makes sense, as UREA is a nitrogen-specific fertilizer and 20-20 has a more balanced but less concentrated phosphorus content.

Although the differences in average phosphorus levels are subtle, they reflect the formulation and purpose of each fertilizer. Fertilizers like 17-17-17 and DAP contribute more effectively to phosphorus levels and are better suited for correcting deficiencies. In contrast, fertilizers like UREA and 20-20 may need to be supplemented with phosphorus-rich products in soils where phosphorus is already low.

How do different combinations of soil and crop types influence Phosphorus levels?

Average of Phosphorous	Column Labels				
Row Labels	Black	Clayey	Loamy	Red	Sandy
Barley	20.42	19.24	18.13	19.10	18.46
Cotton	18.71	19.32	19.60	18.01	19.32
Ground Nuts	17.59	17.08	19.01	21.32	17.42
Maize	18.17	18.02	17.87	17.79	17.07
Millet	16.91	16.98	19.72	16.53	19.02
Oil seeds	19.54	16.44	17.49	17.16	18.01
Paddy	19.60	17.63	17.63	19.37	19.43
Pulses	18.57	17.84	19.37	19.52	19.60
Sugarcane	18.85	18.61	19.55	18.46	16.79
Tobacco	17.29	19.85	17.87	19.26	19.30
Wheat	19.75	18.25	17.08	20.25	19.41



- **average Phosphorus levels across different crop and soil type combinations**

Phosphorus levels vary more by crop than by soil type. Across almost all crops, phosphorus values range roughly from 16.4 to 21.3, indicating moderate variability.

Barley shows the highest phosphorus in black soil (20.42%), indicating that black soil may naturally support phosphorus-rich barley fields.

Ground Nuts has the highest phosphorus in red soil (21.32%), suggesting red soil may be particularly nutrient-enriched or intensively fertilized for this crop.

Wheat and Paddy both show high phosphorus values in sandy soil (19.4%), which is interesting because sandy soils typically have low nutrient retention this may indicate strong fertilization practices.

Oil Seeds show the lowest value (16.44%) in clayey soil, signaling a possible nutrient mismatch.

For soils

Black soil provides higher phosphorus levels for crops like Barley, Oil Seeds, and Wheat.

Clayey soil shows the most variability — sometimes supporting high levels (Tobacco = 19.85%) and sometimes the lowest (Oil Seeds = 16.44%).

Sandy soil, despite its poor nutrient-holding capacity, provides surprisingly high phosphorus for Tobacco and Wheat, possibly due to compensatory fertilization.

The phosphorus content across crop–soil combinations shows meaningful variation, revealing how both natural soil characteristics and fertilization practices interact. Crops like Barley, Ground Nuts, and Wheat perform best in black, red, or sandy soils, where phosphorus levels are higher. In contrast, Millets and Oil Seeds consistently show lower phosphorus levels across most soils.

This suggests a need for:

- **Targeted phosphorus management based on both crop type and soil,**
- **Soil testing before planting, and**
- **Adjusting phosphorus input where specific crops (like oil seeds or millets) fall below optimal levels.**

V. Global Conclusion

This analysis offers a comprehensive overview of the relationships between soil characteristics, crop types, nutrient levels, and fertilizer usage. Across the dataset, most environmental and chemical variables such as temperature, humidity, and moisture are relatively stable, indicating

that the samples likely originate from a geographically consistent and well-managed agricultural zone.

However, deeper analysis reveals meaningful differences in nutrient distribution, especially for nitrogen, phosphorus, and potassium, which are essential for healthy crop development. Most soils show low potassium and phosphorus levels, highlighting the need for targeted fertilization strategies. Nitrogen distribution, while more varied, also suggests localized practices or crop-specific demands (e.g., maize and cotton).

Cross-analysis of soil–crop–fertilizer combinations demonstrates that:

- Some crops, like barley and wheat, thrive better in nutrient-rich soils such as black or clayey.
- Crops like millets and oil seeds are often associated with lower nutrient levels, possibly due to their natural tolerance or lower input systems.
- Fertilizers such as DAP and 17-17-17 are more commonly applied in nutrient-deficient areas and offer more balanced nutritional support.

The consistency in averages, paired with the variability in distribution, suggests that fertilizer practices are somewhat standardized, but nutrient availability varies—a gap that could be improved through precision agriculture and soil-specific fertilization plans.

- **How this report can help?**

This report can help farmers and agricultural advisors to:

- Choose the most suitable fertilizer based on both crop type and soil condition.
- Avoid over- or under-application by matching actual nutrient needs.
- Identify crops that are at risk due to nutrient deficiencies, and apply corrective strategies.
- Promote sustainable farming practices by minimizing input waste and optimizing yield potential.

With the right guidance, these insights can support more efficient, climate-resilient, and data-driven farming.